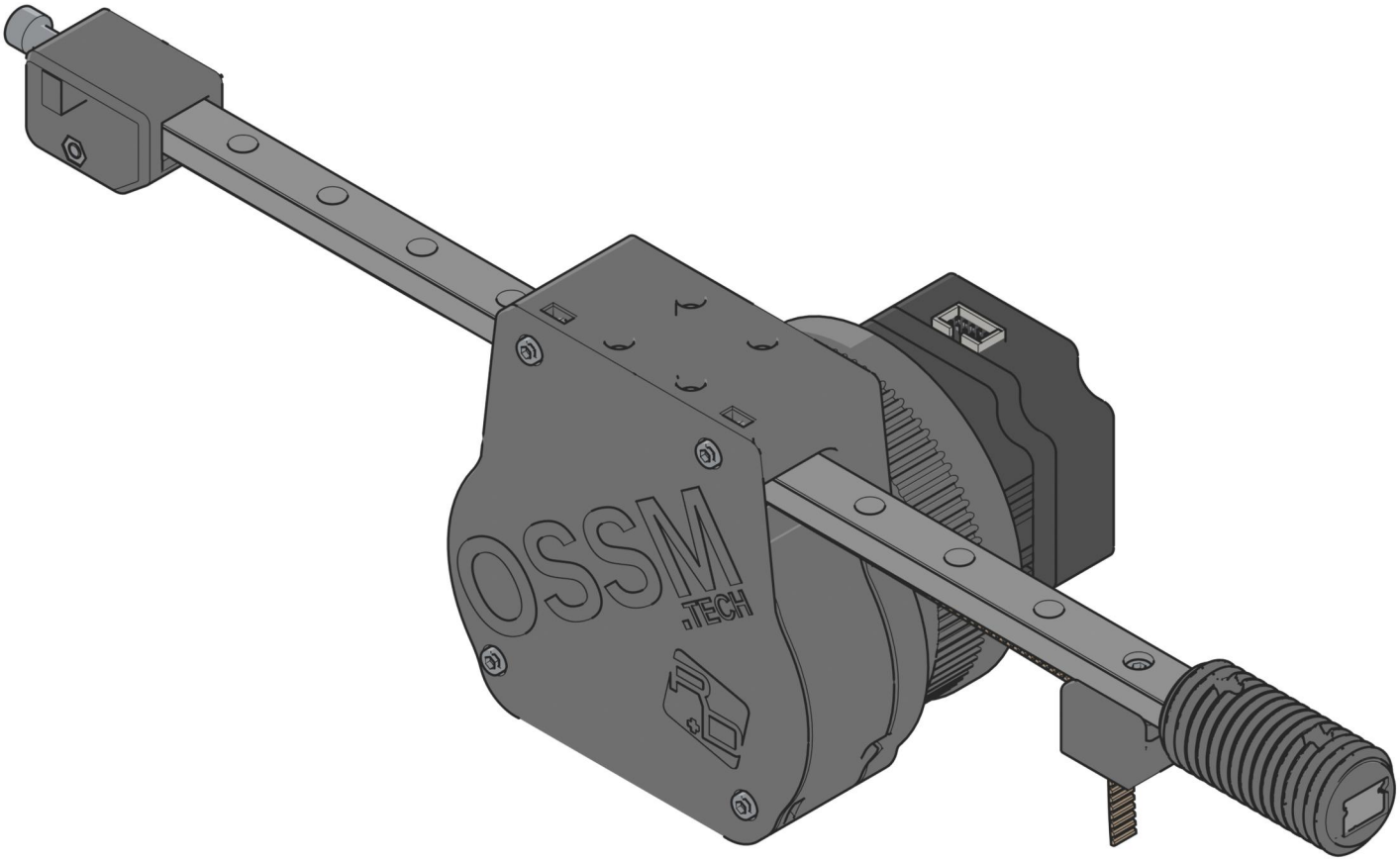
OSSM

RESEARCH + DESIRE

Actuator Build Guide

Actuator · Version 1.0.17



Support
support@researchanddesire.com

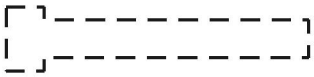
Assembly instructions
ohai.researchanddesire.com/ossm

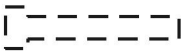
Actual size

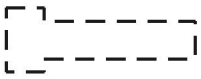


Print at 100 % (actual size). Lay a part on its dashed outline to identify it.

Bolts


M5×35 Screw


M3×20 Screw


M5×20 Screw


M3×8 Screw

Nuts


M5 Nut


M3 Nut

Hex keys

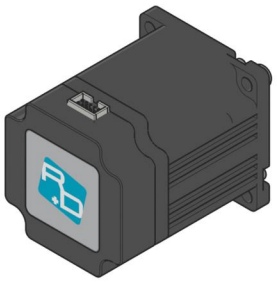

4 mm Hex key


2.5 mm Hex key

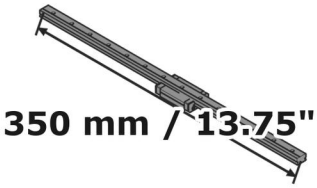

2 mm Hex key

Parts

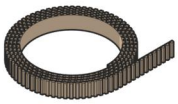
Motor and linear rail



1×
Servo motor

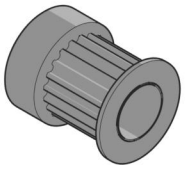


1×
Linear rail



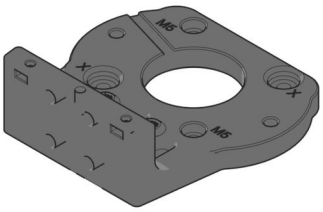
500 mm

1×
Belt

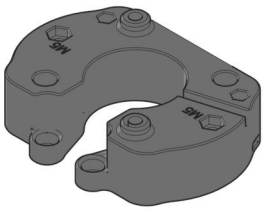


10 mm
1×
Pulley

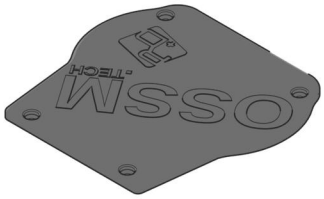
3D printed



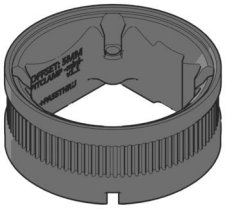
1×
Body bottom



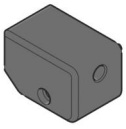
1×
Body middle



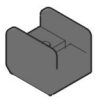
1×
Cover



1×
Motor ring



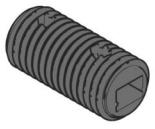
1×
Belt tensioner



1×
Belt clamp bottom



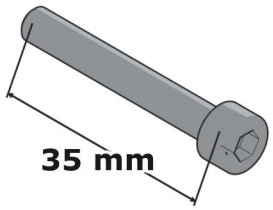
1×
Belt clamp top



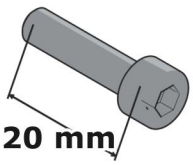
1×
24 mm
Threaded end effector

Parts

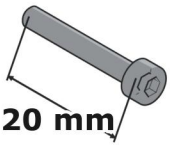
Bolts



4×
M5×35 Screw



3×
M5×20 Screw



2×
M3×20 Screw



10×
M3×8 Screw

Nuts



5×
M5 Nut



8×
M3 Nut

Bearings



6×
5×11×4
Bearing

Tools

Required



4 mm Hex key

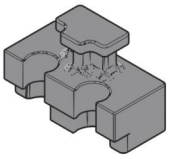


2.5 mm
Hex key



2 mm Hex key

3D printed jigs

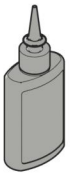


Bearing jig



3.8 mm
Pulley spacer

Recommended

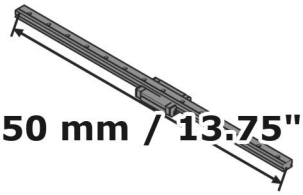


Threadlocker
(blue 243)

Legend

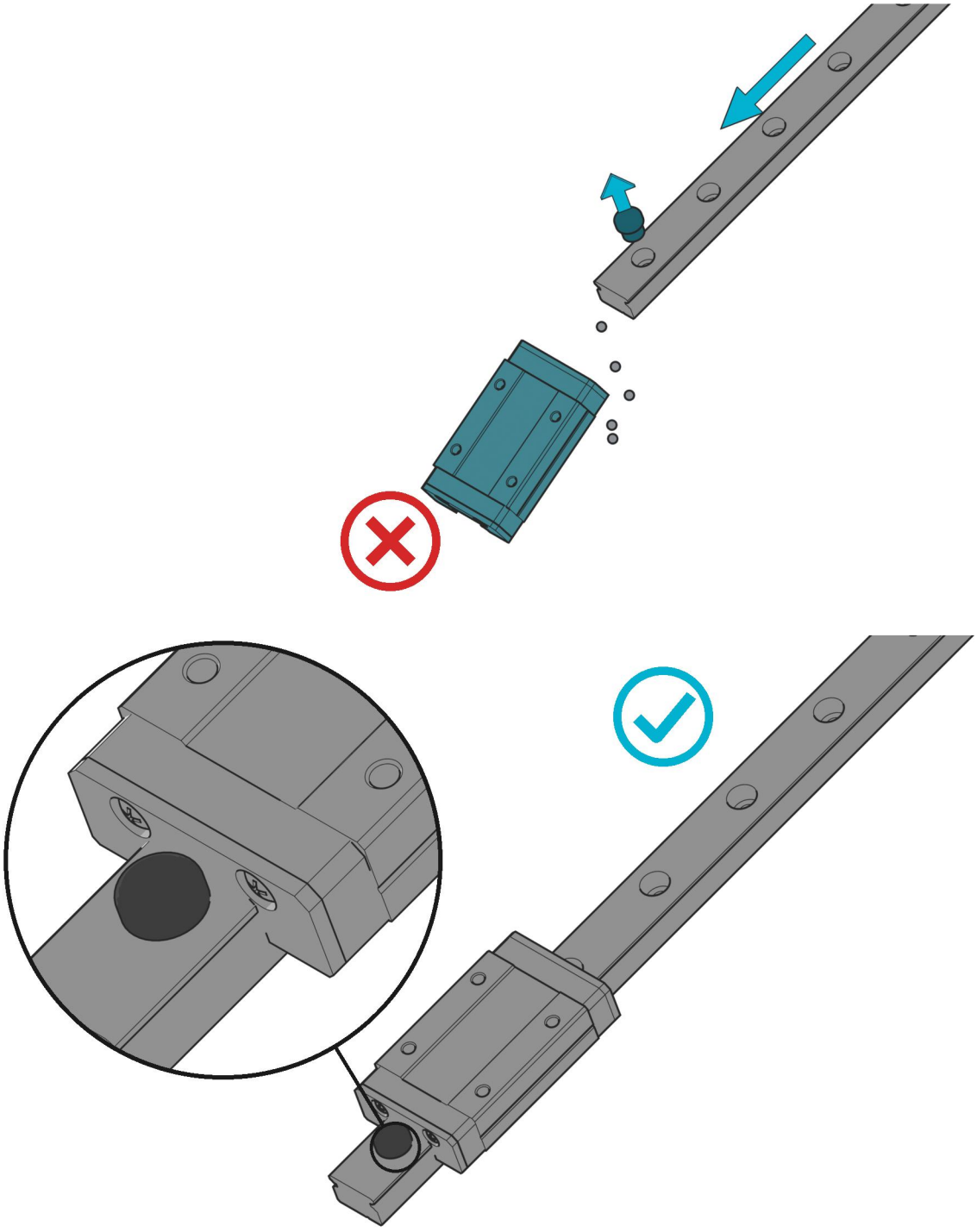
In a step's parts box, this drop beside a screw's quantity means:
put threadlocker (blue 243) on the screw's thread before it goes in.

Warnings ⚠



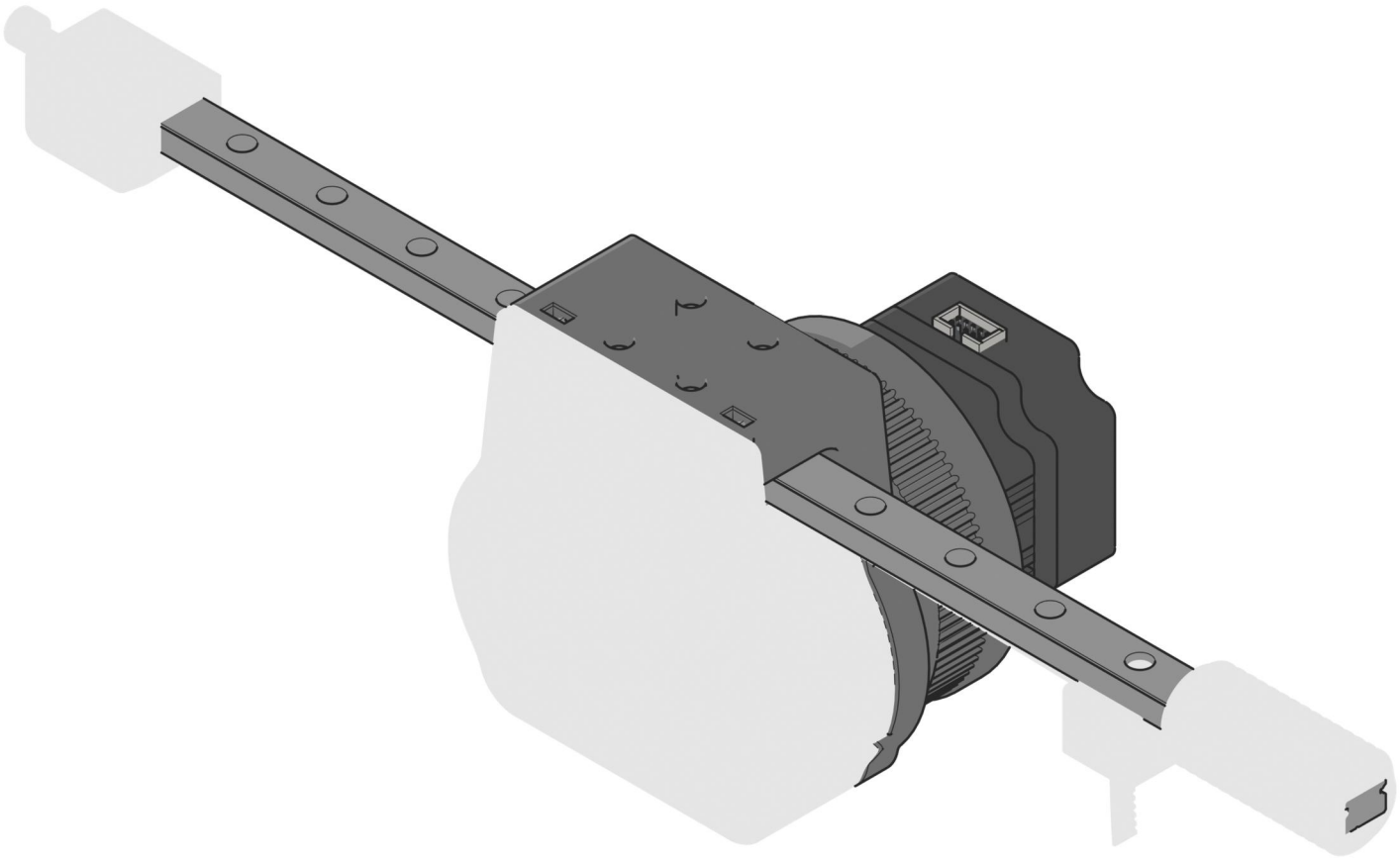
1×

Linear rail

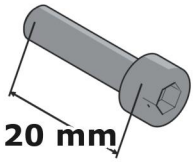


⚠ **Never let the carriage slide off the rail: its balls fall out.**

1 The motor head

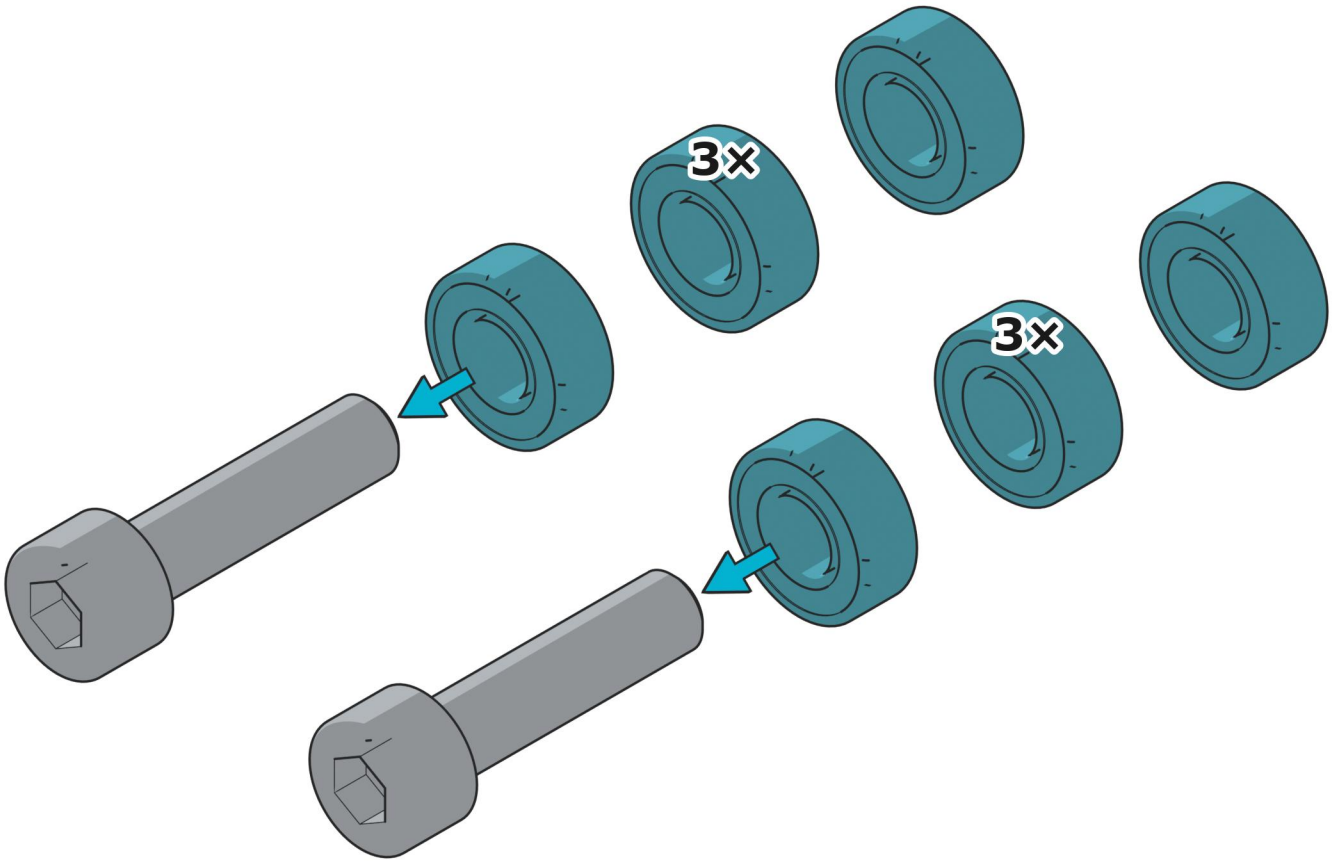


1

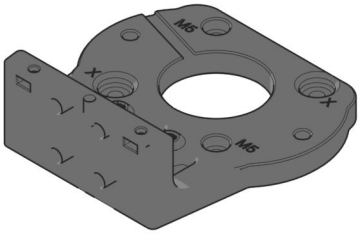


2×
M5×20 Screw

6×
5×11×4 Bearing



2

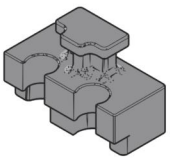


1×

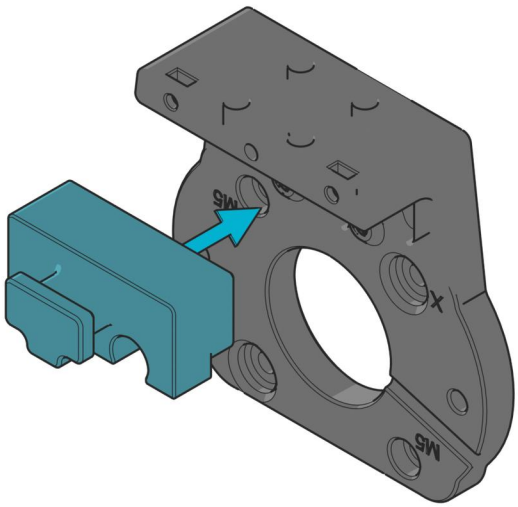
Body bottom

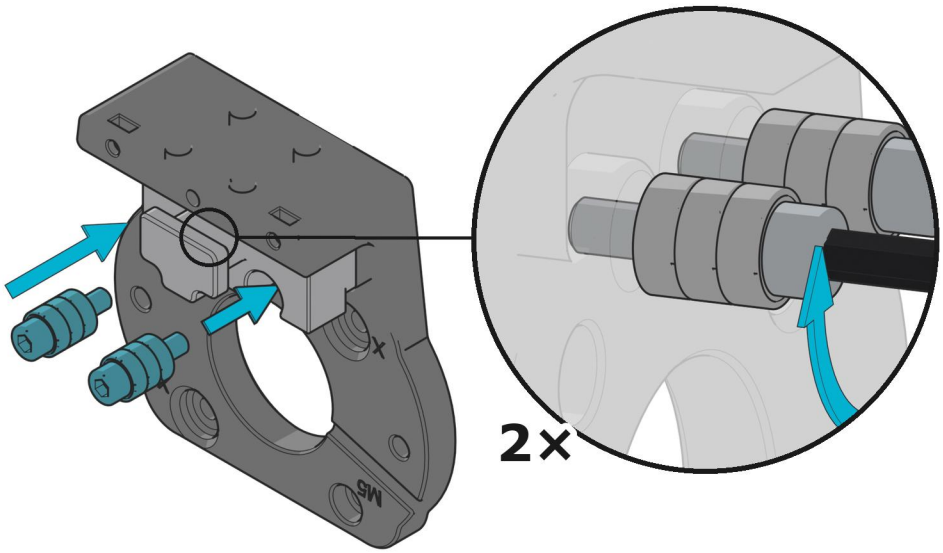


4 mm Hex key



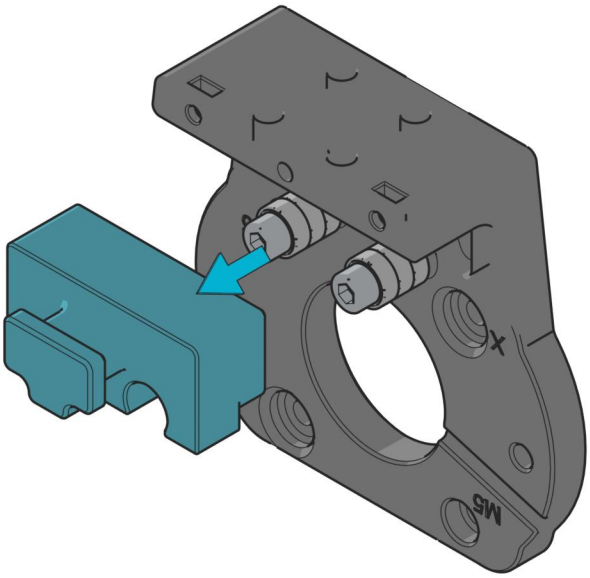
Bearing jig

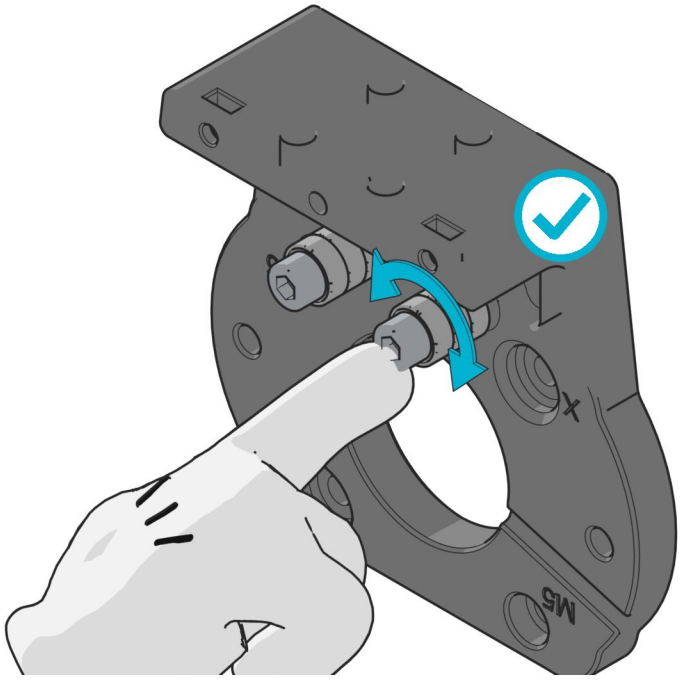




Snug only: do not over-tighten the idler screws.

3





Snug only: each bearing must spin freely. If one does not, loosen its screw a little.

4

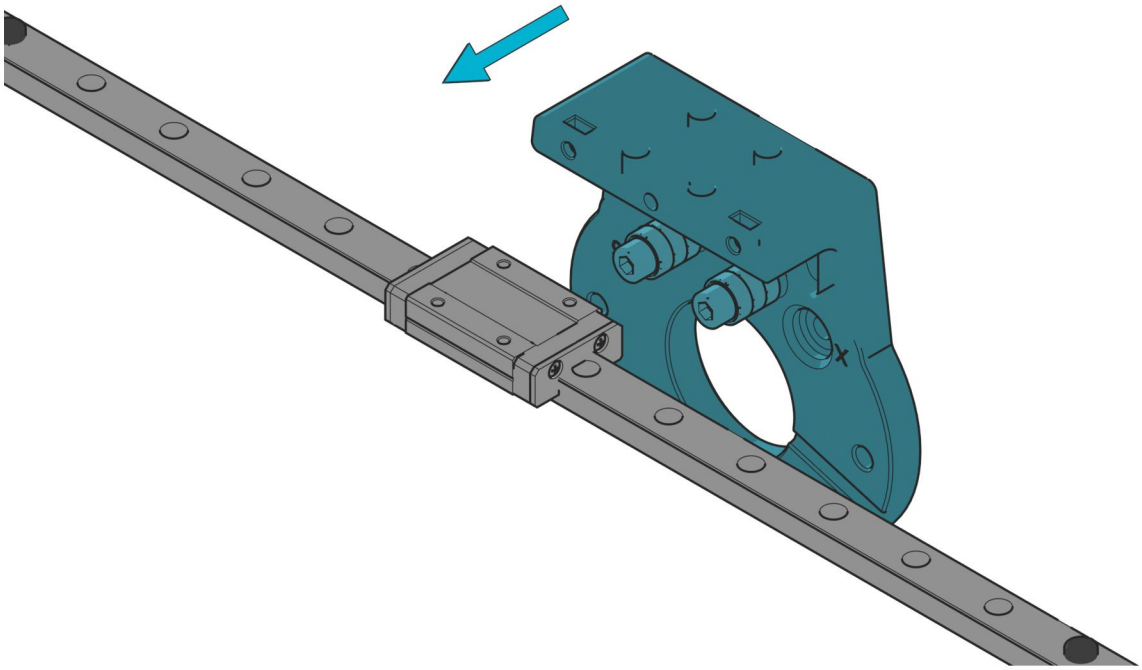


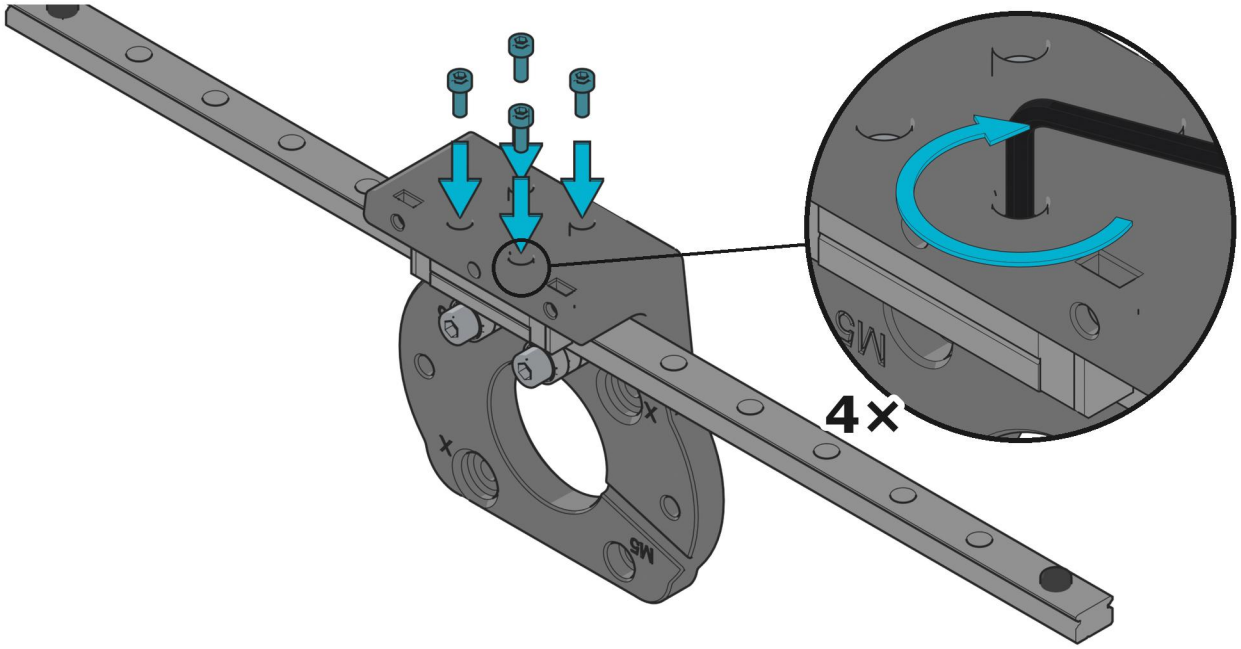
4× M3×8 Screw

1× Linear rail

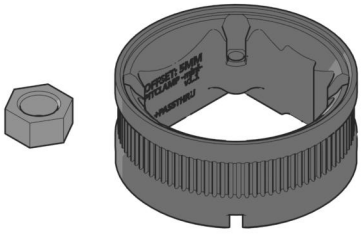


2.5 mm Hex key



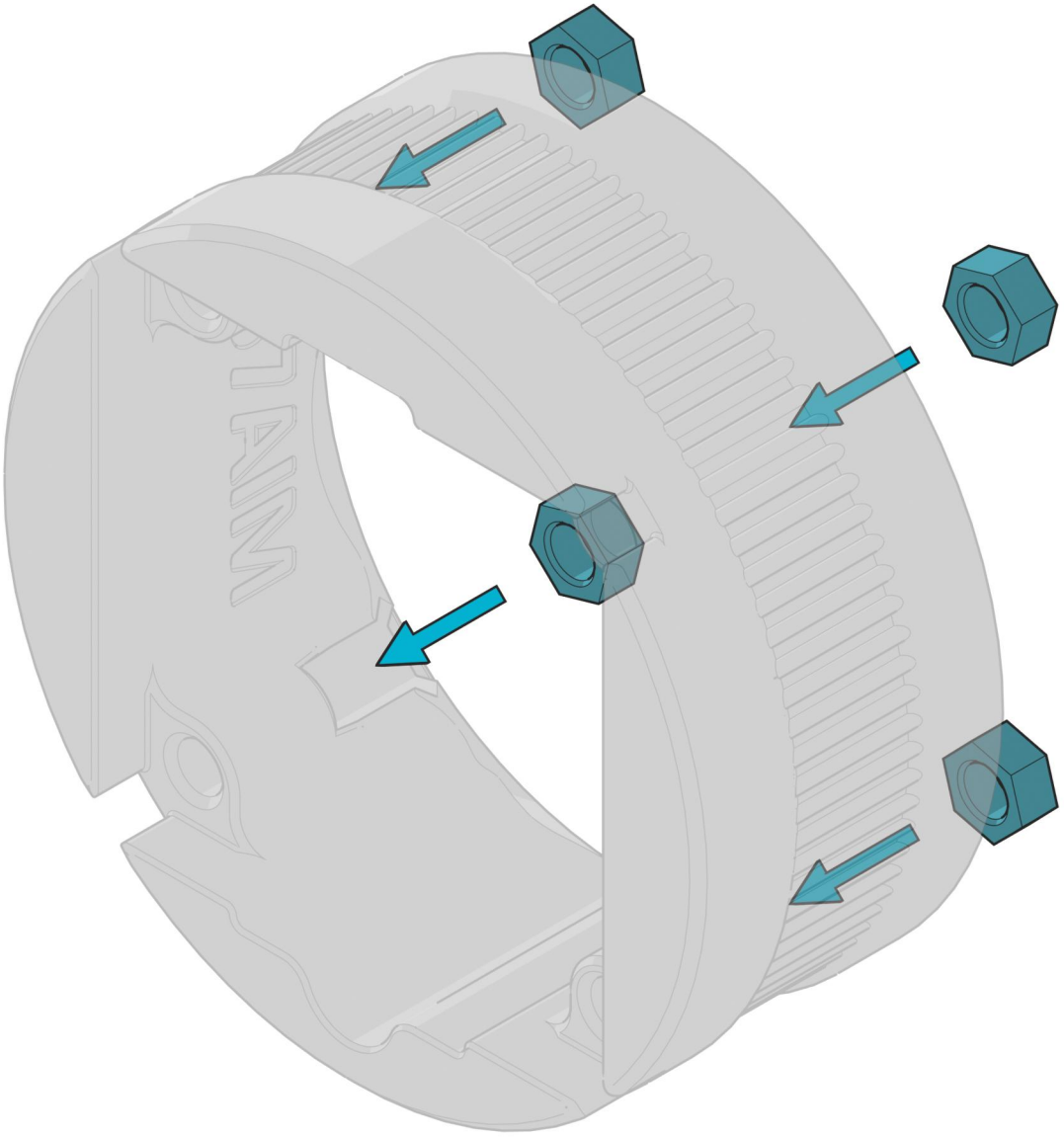


5



4× M5 Nut

1× Motor ring

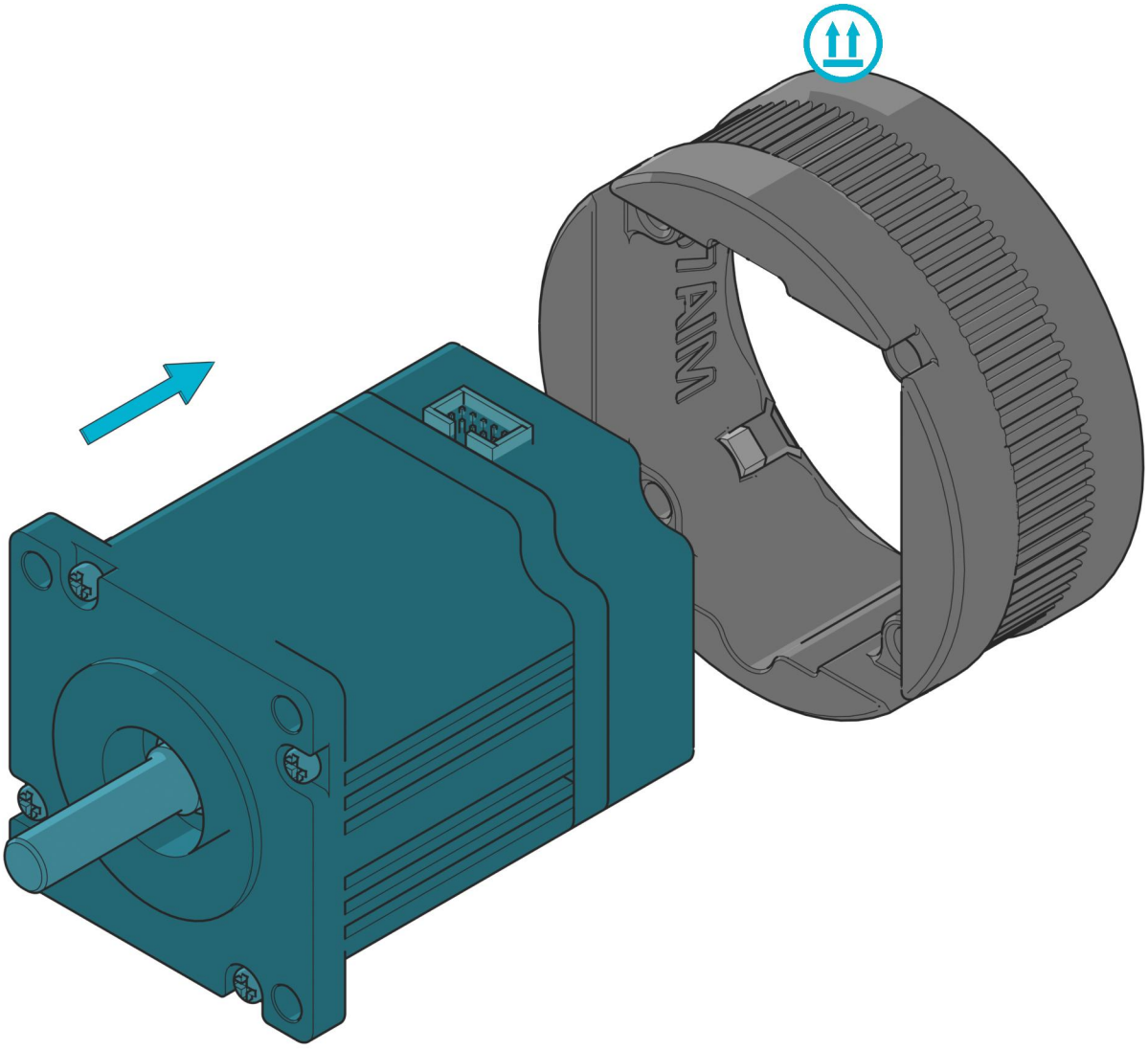


6

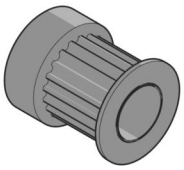


1×

Servo motor



7



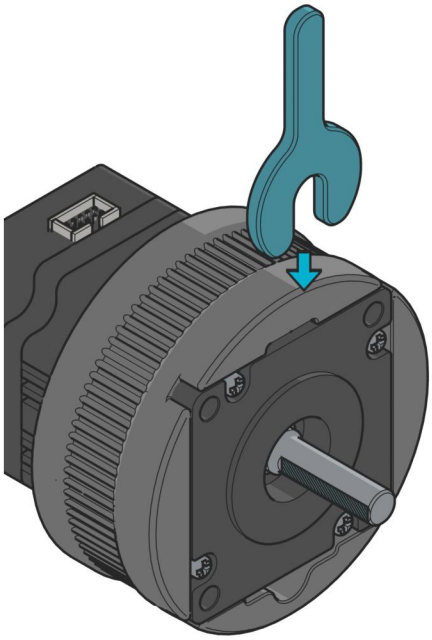
10 mm

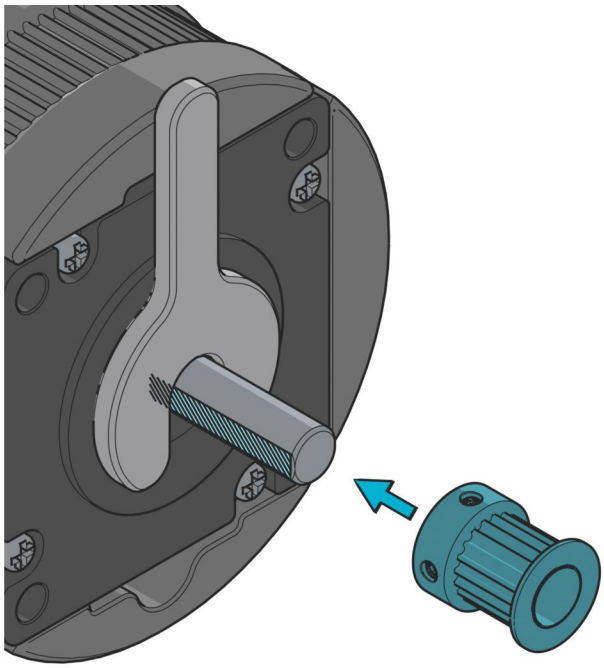
1×

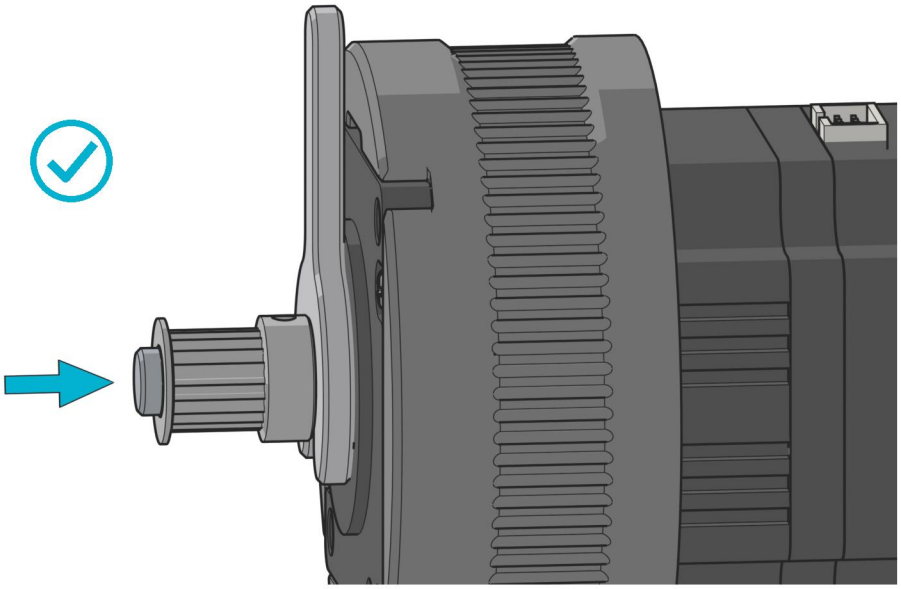
Pulley



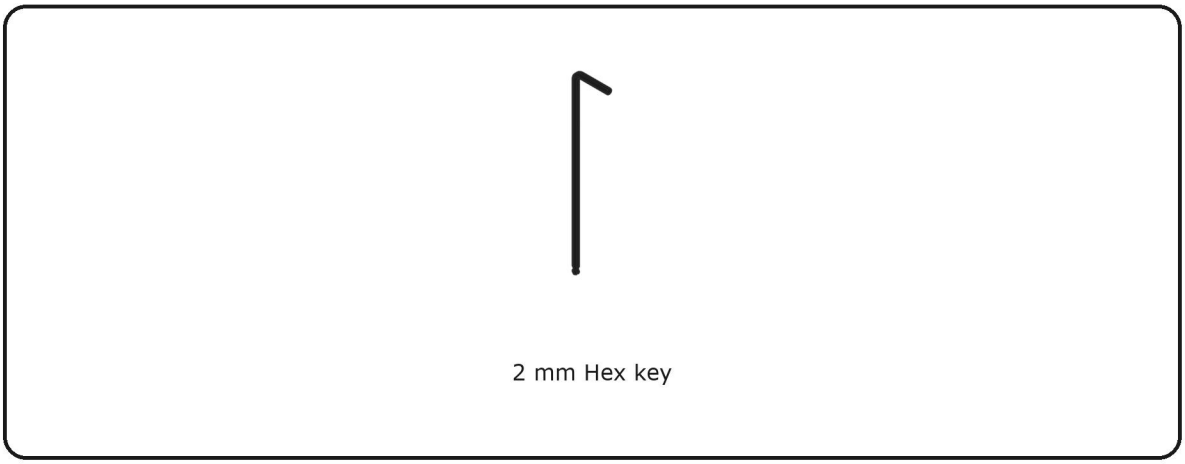
3.8 mm
Pulley spacer

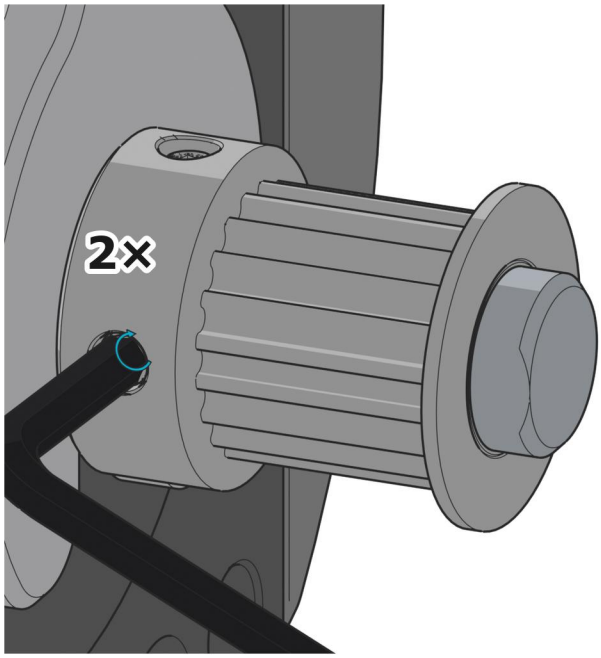


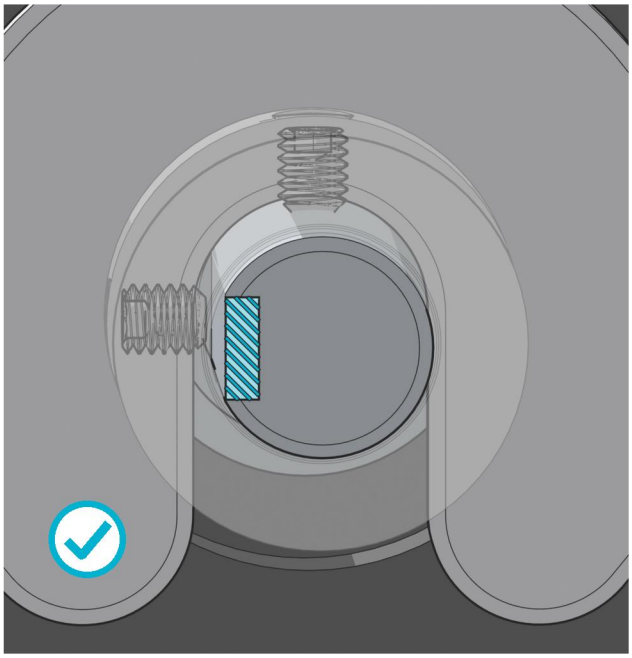


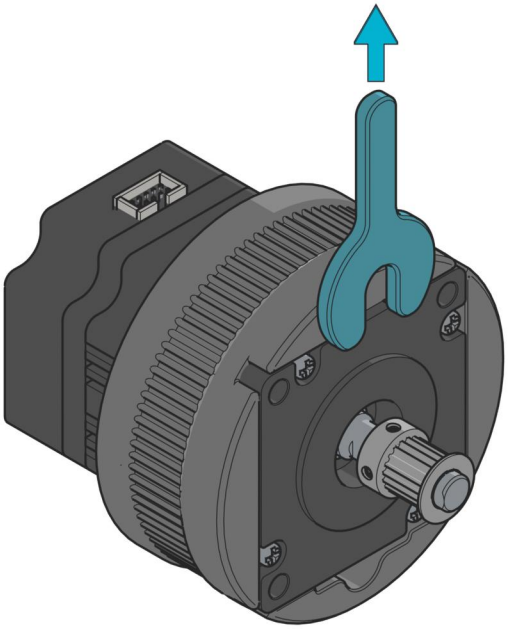


8



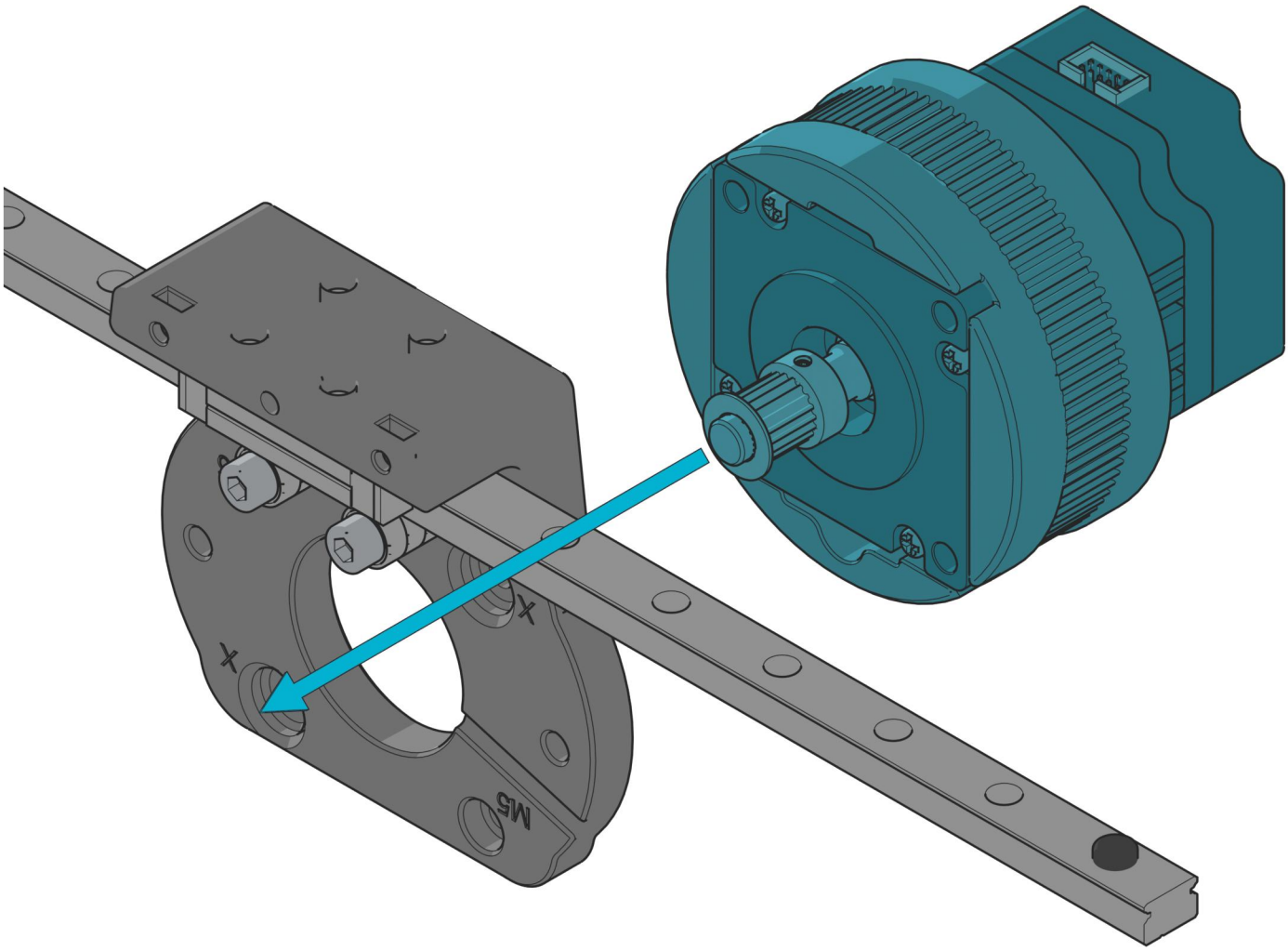




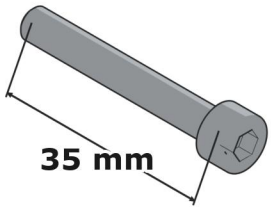


Keep the pulley pushed against the spacer while you tighten.

9



10

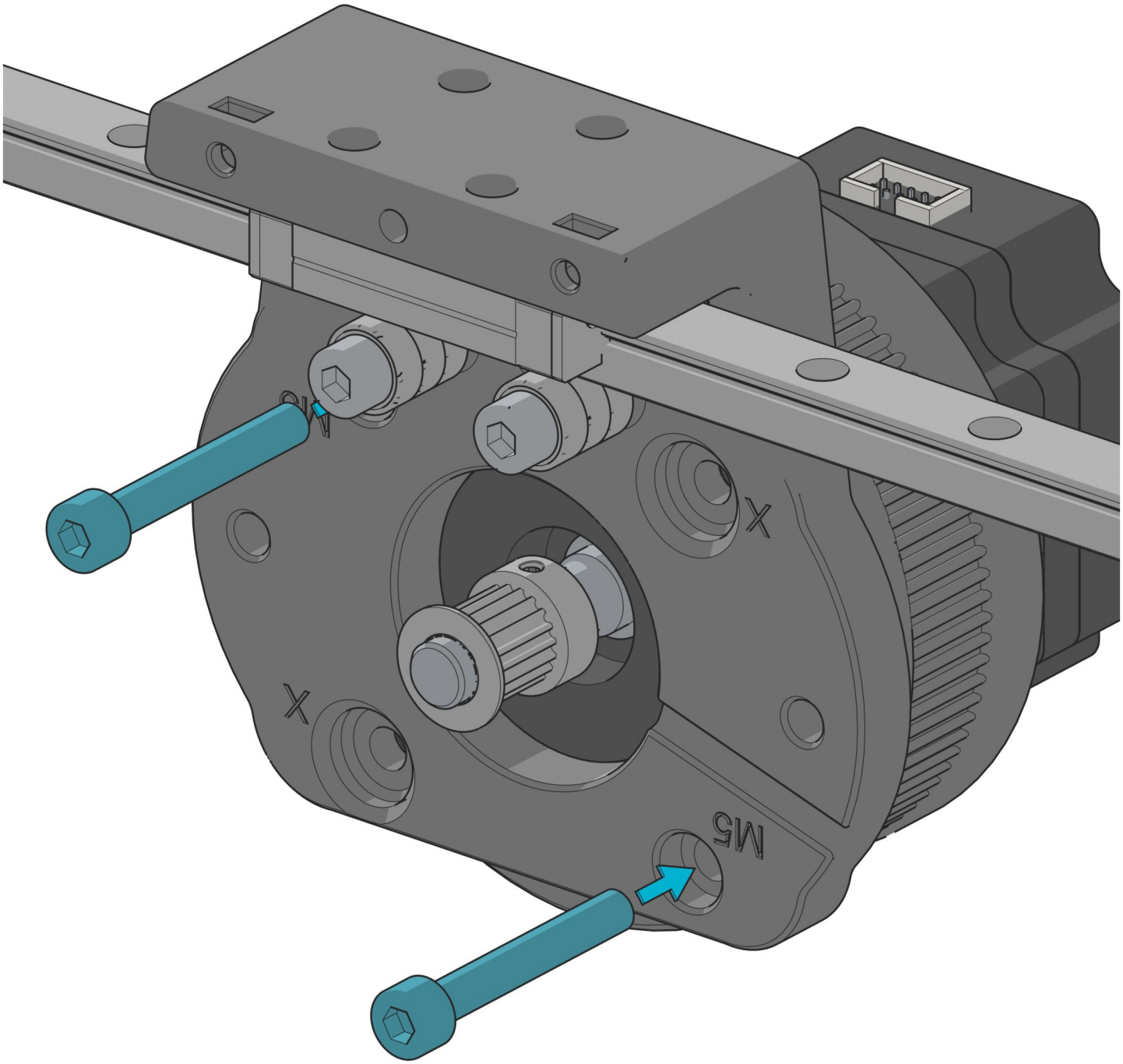


2×

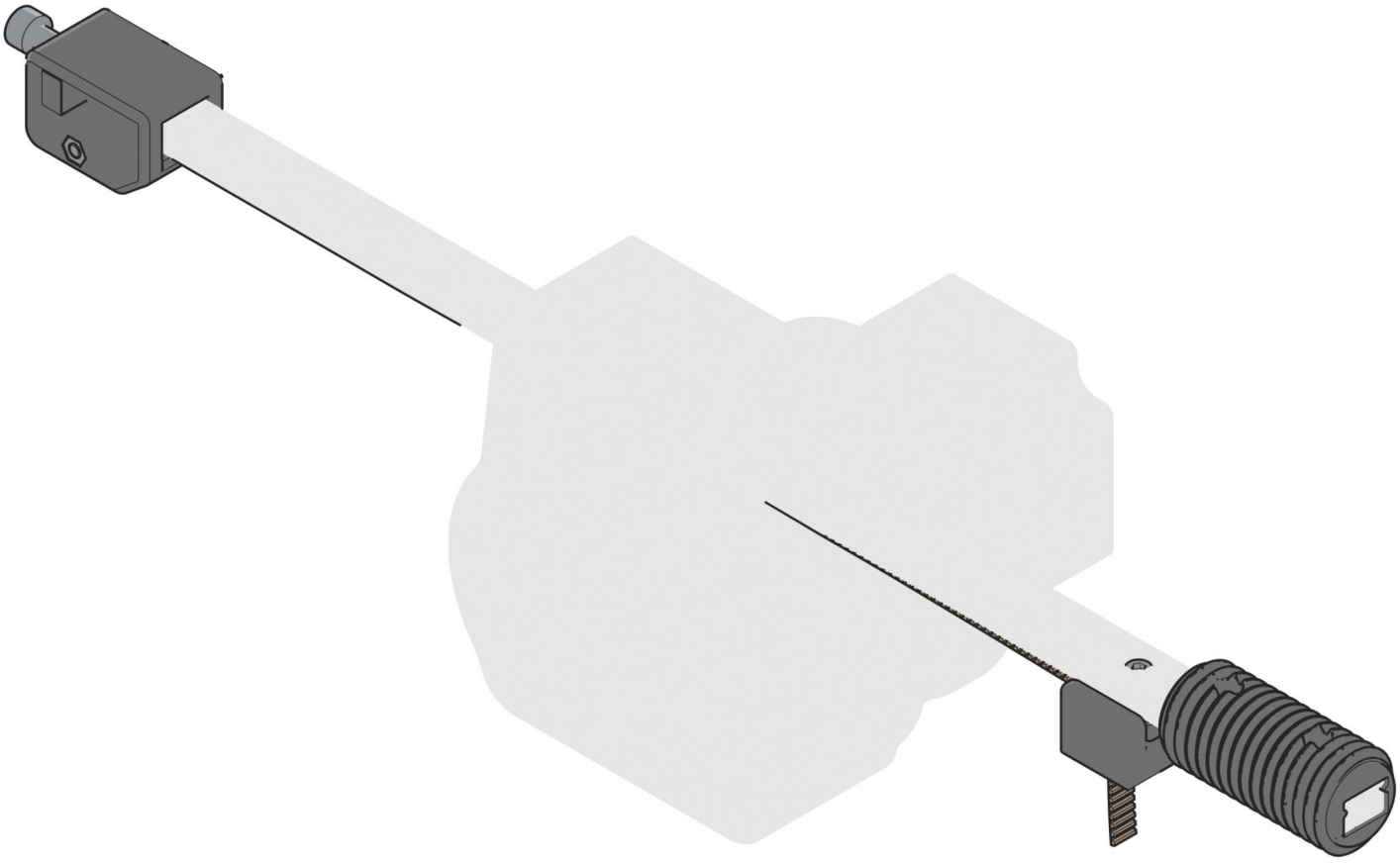
M5×35 Screw



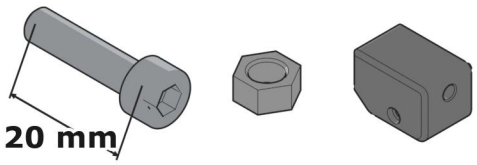
4 mm Hex key



2 The belt



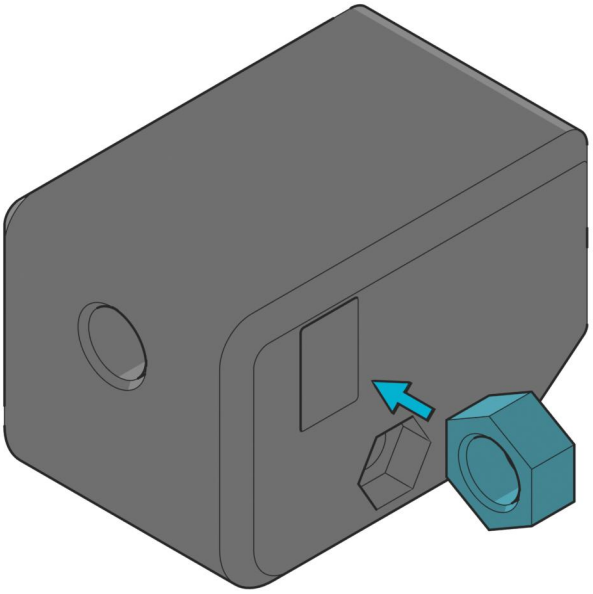
11

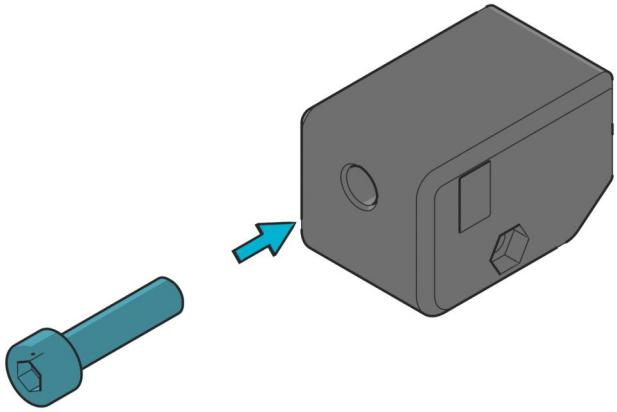


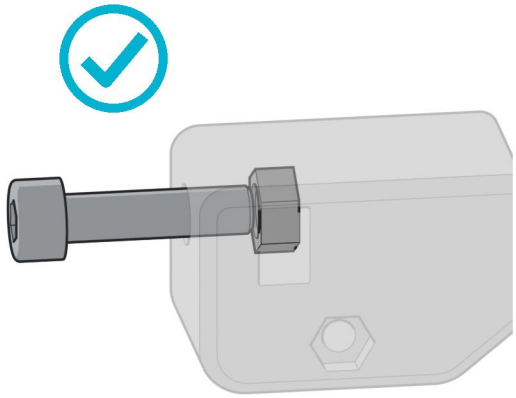
1× M5×20 Screw

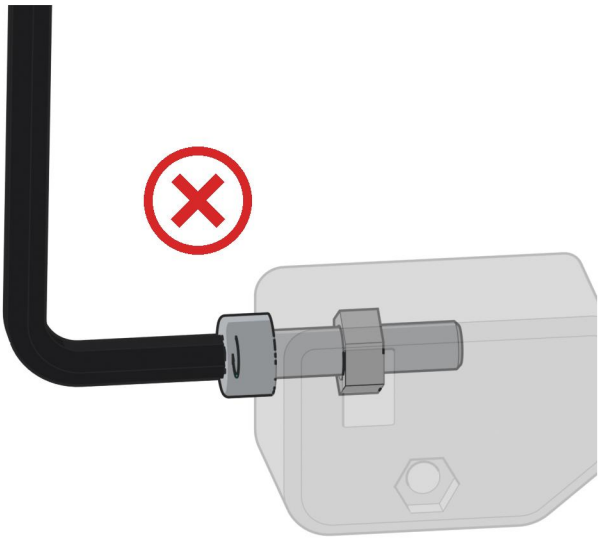
1× M5 Nut

1× Belt tensioner

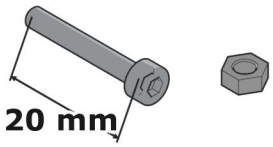








12

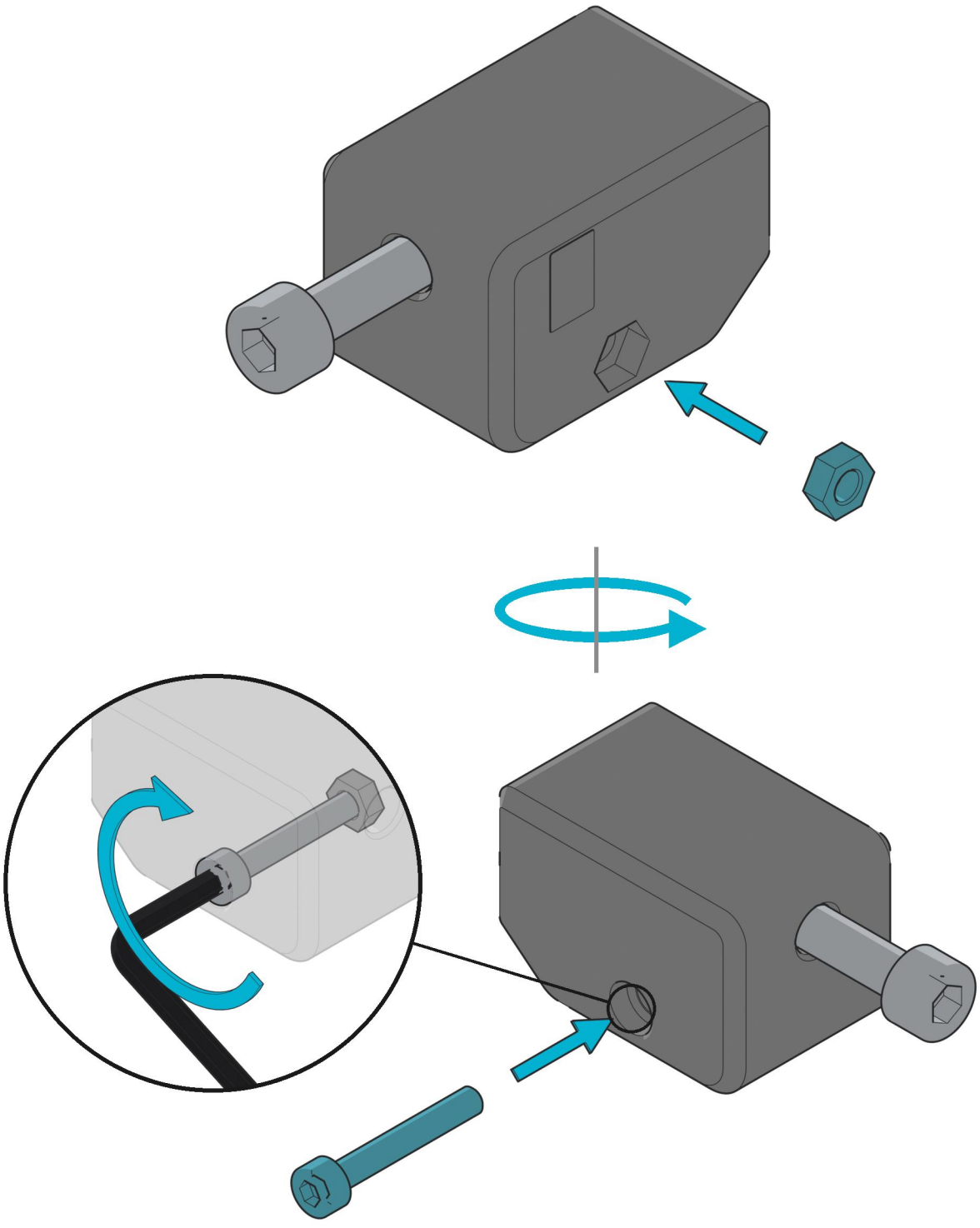


1× M3×20 Screw

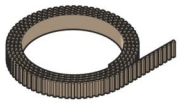
1× M3 Nut



2.5 mm Hex key



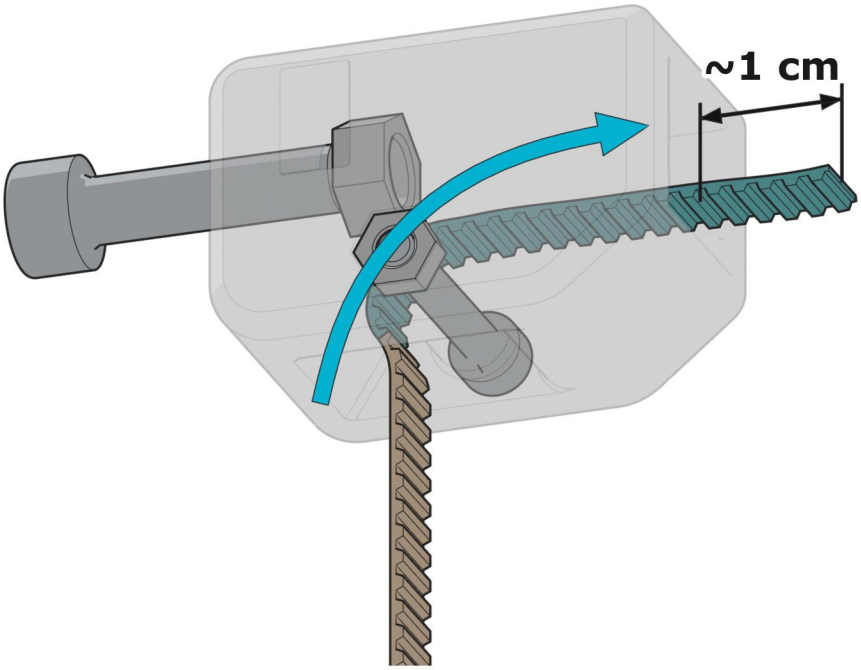
13

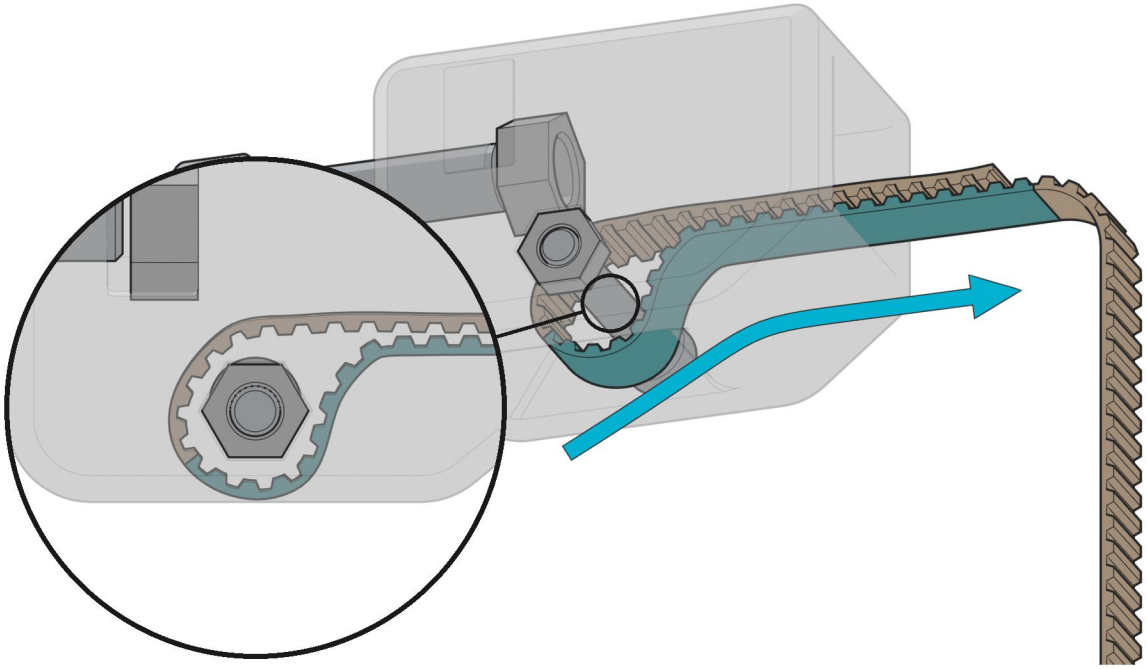


500 mm

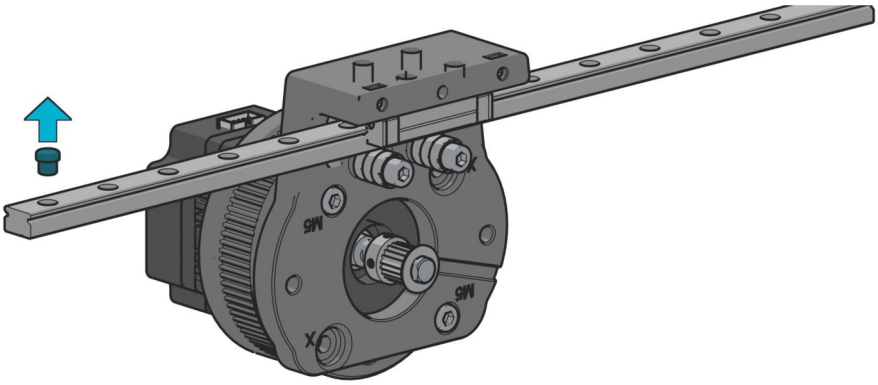
1×

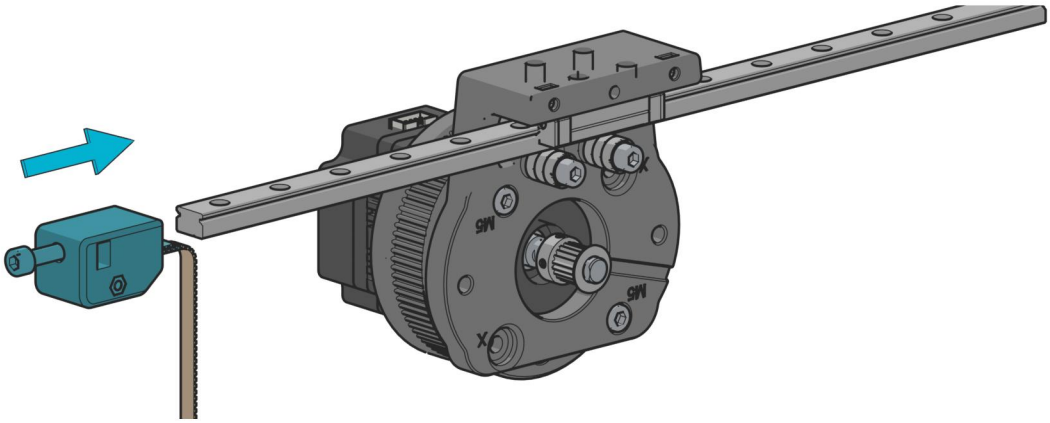
Belt

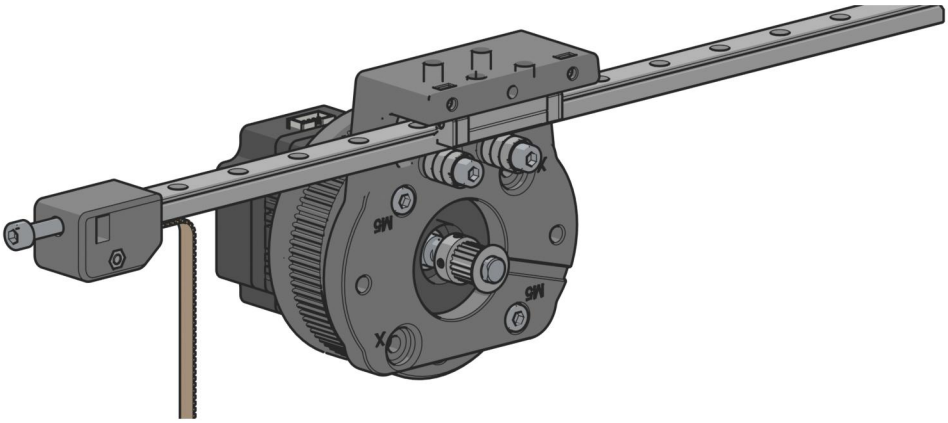




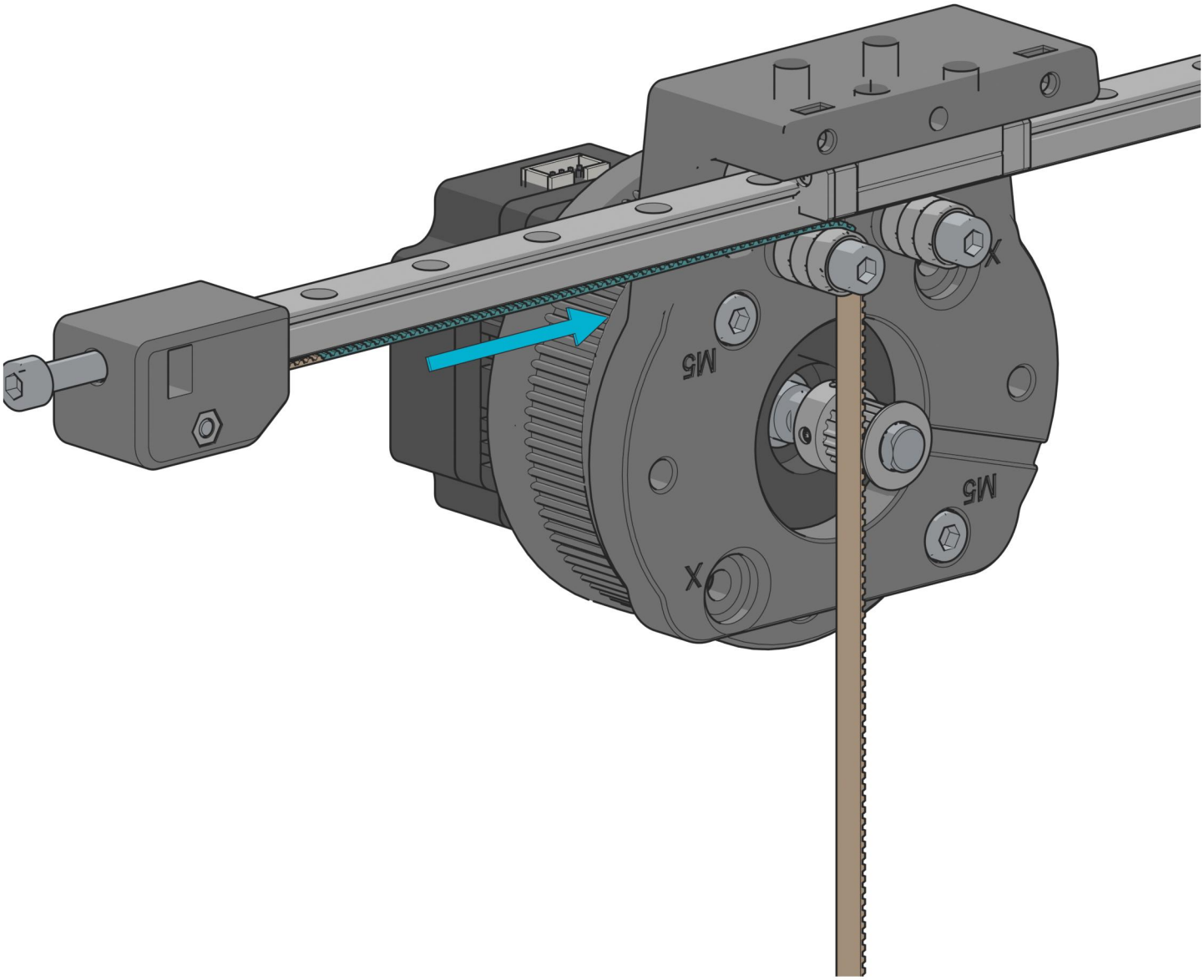
14



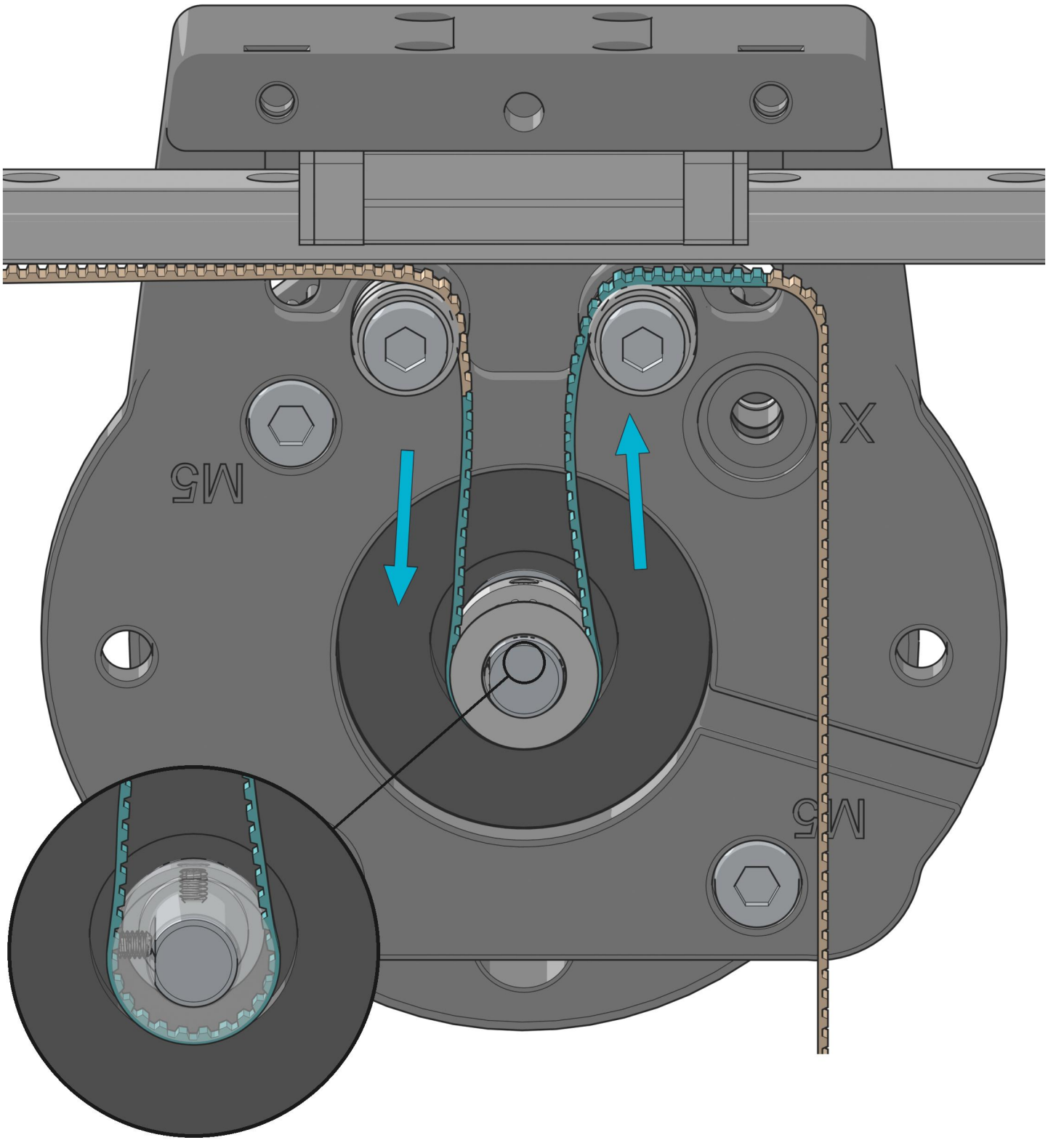




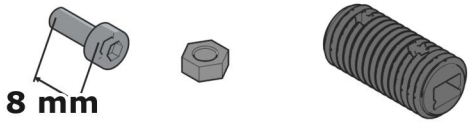
15



16



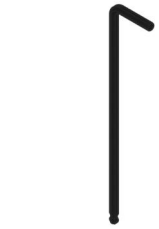
17



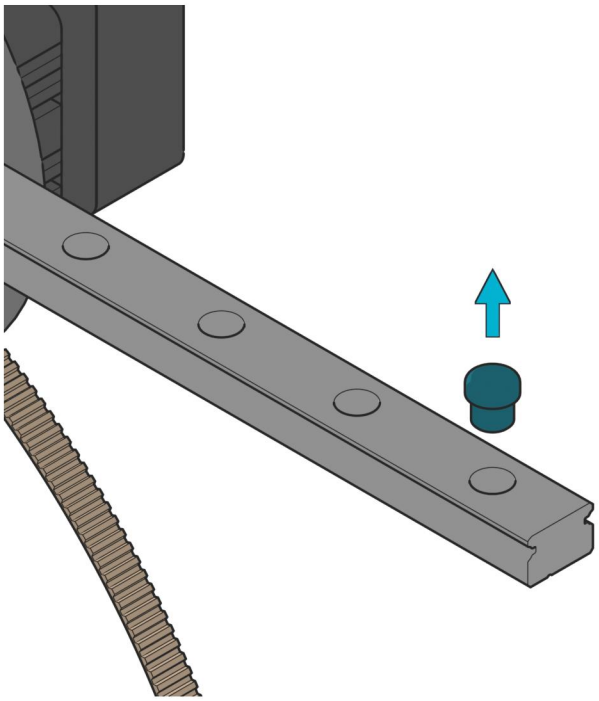
2× M3×8 Screw

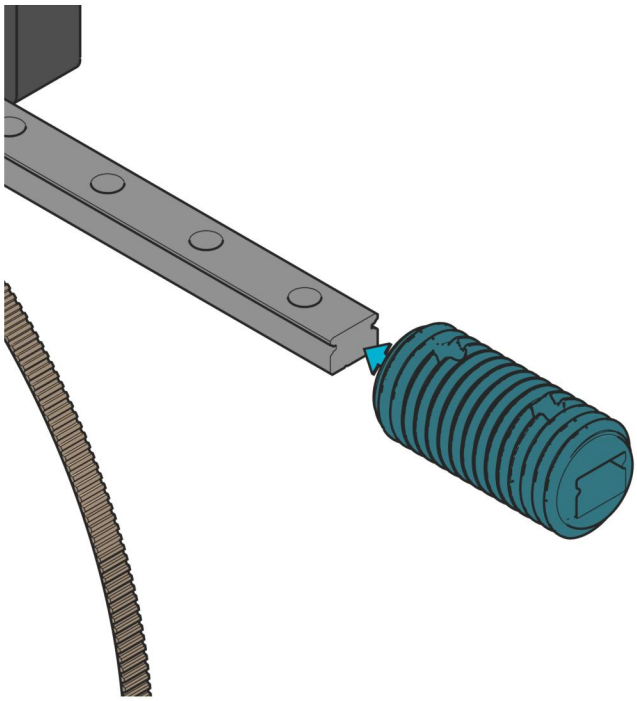
2× M3 Nut

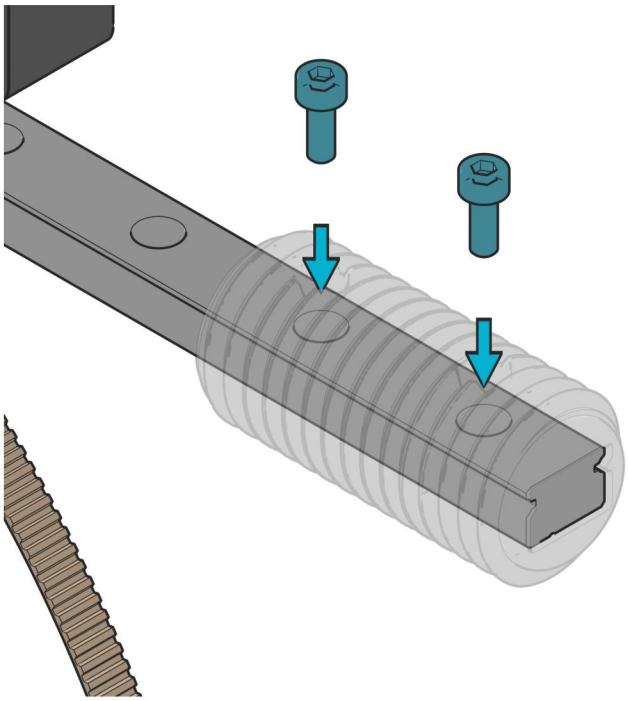
1× 24 mm Threaded end effector

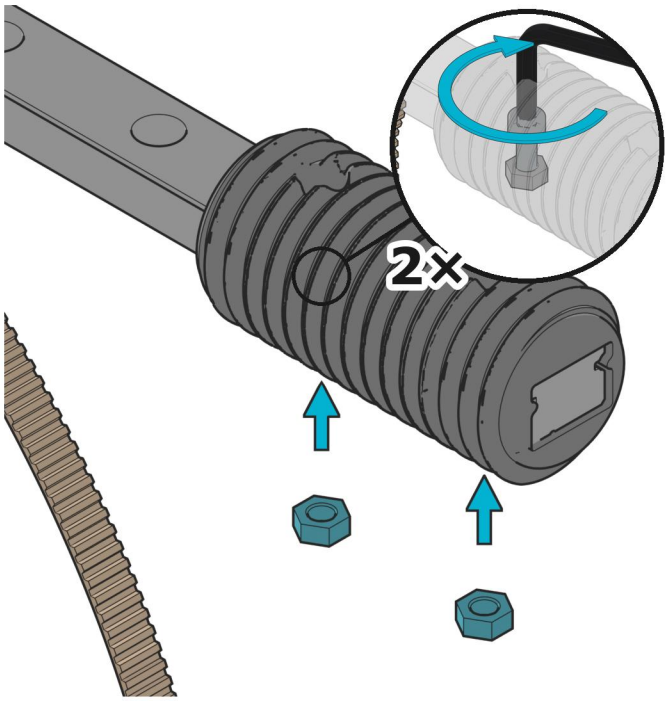


2.5 mm Hex key

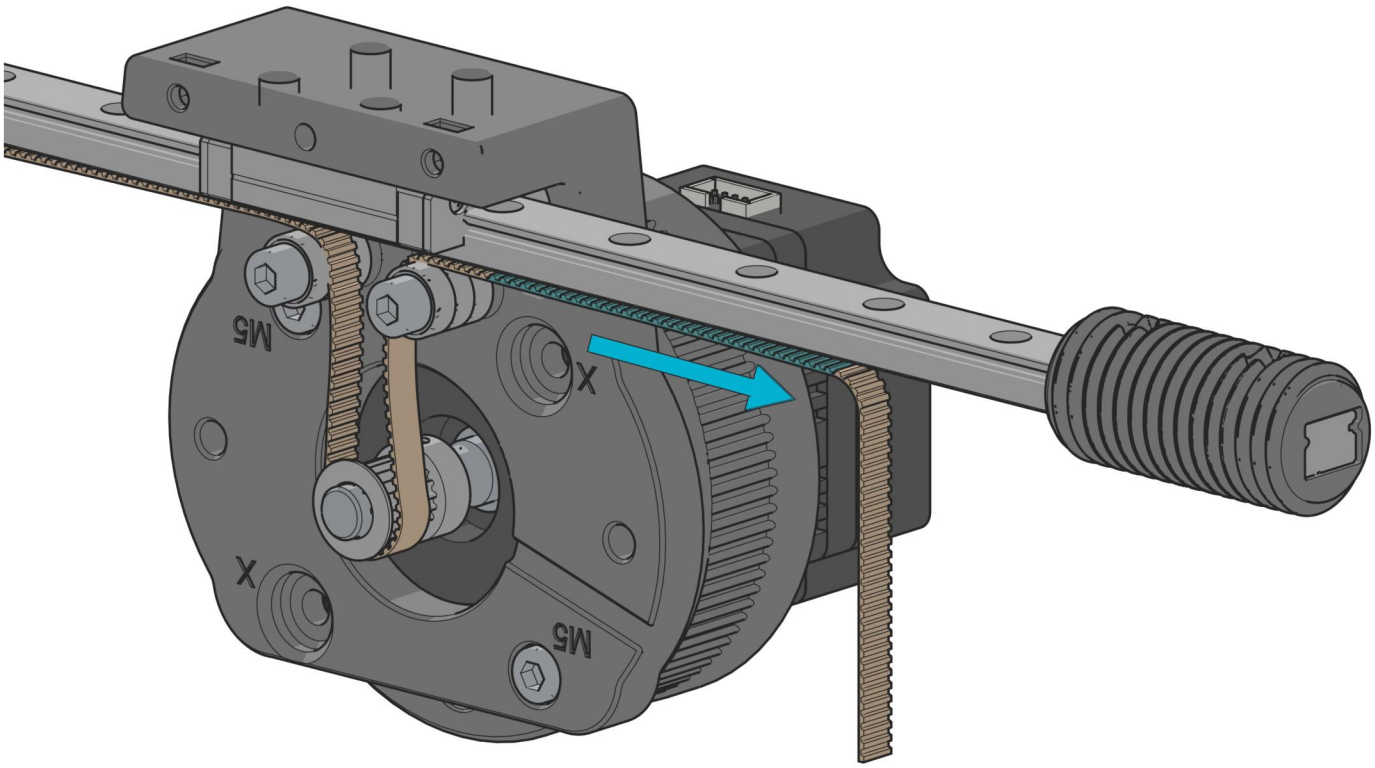








18

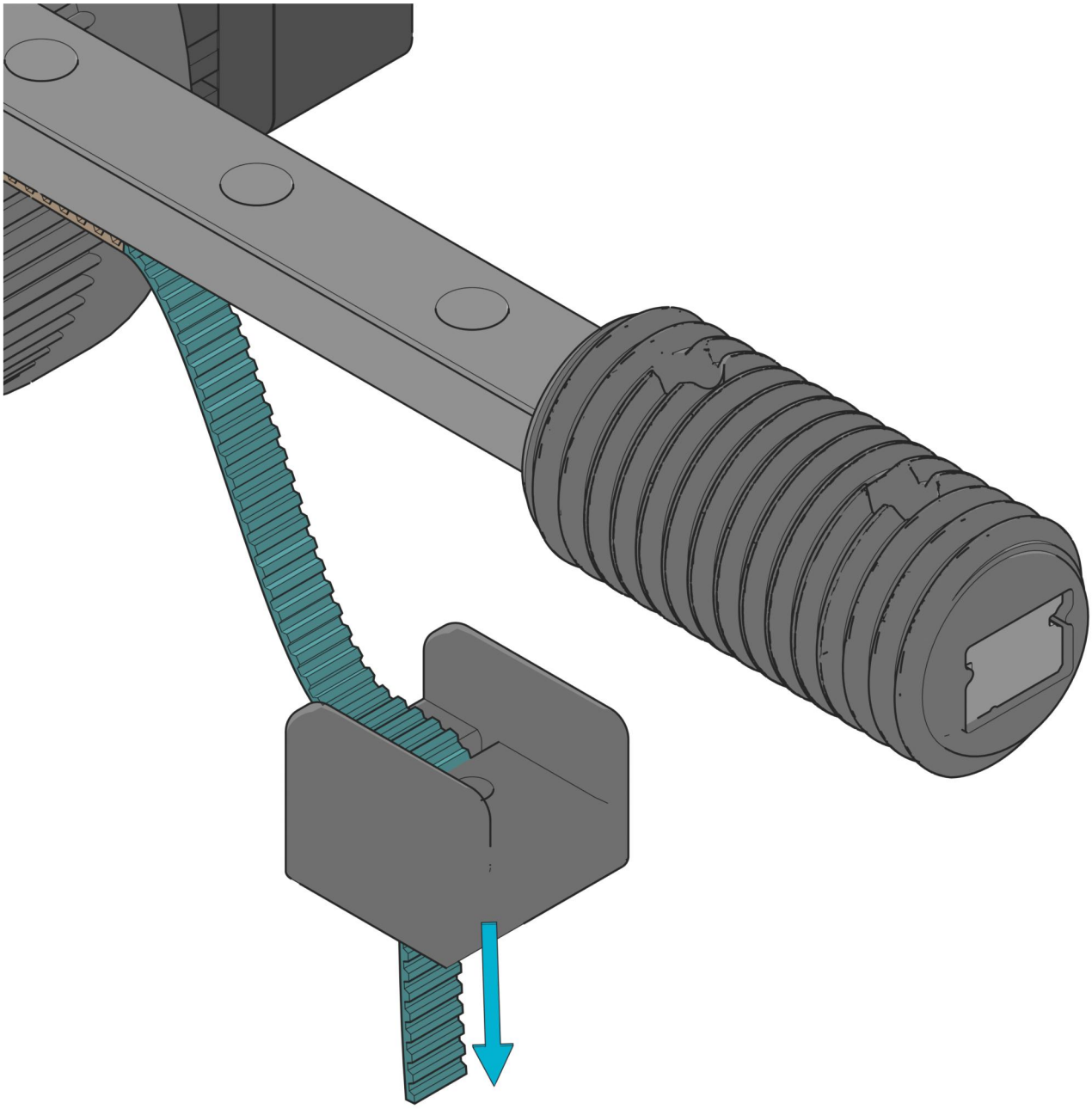


19



1×

Belt clamp
bottom

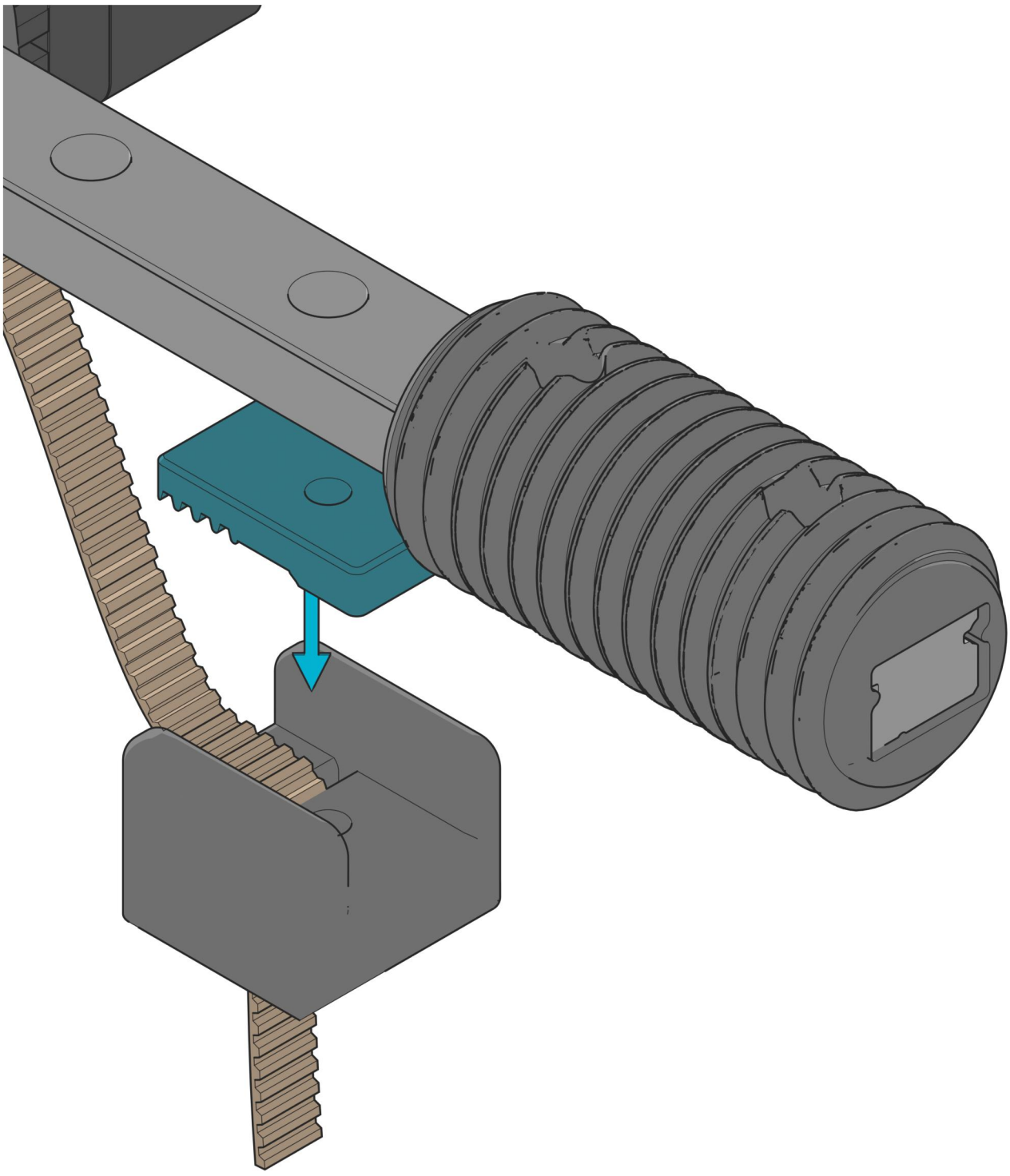


20

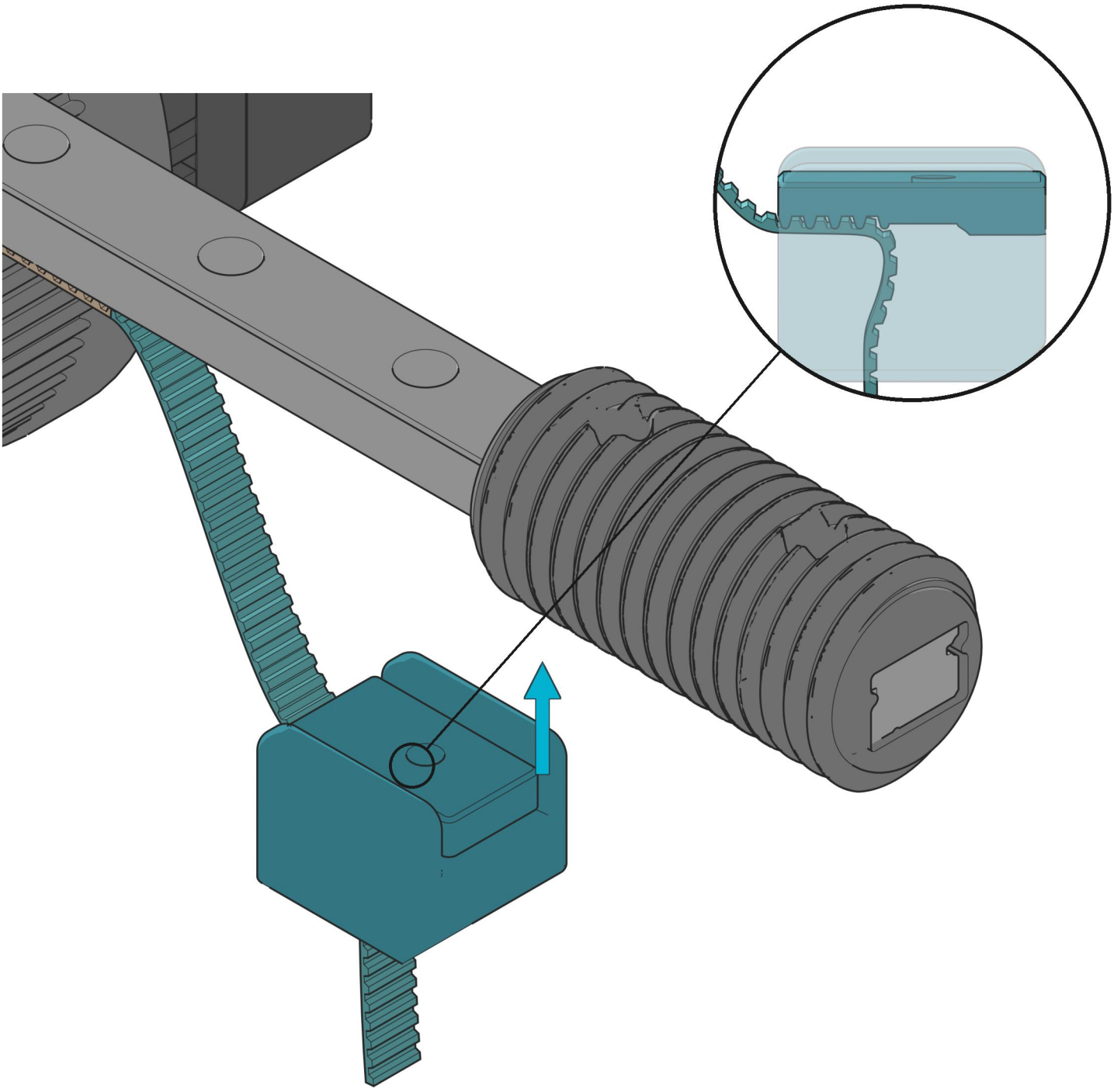


1×

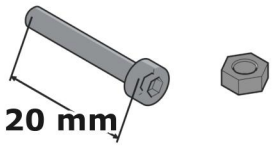
Belt clamp top



21



22

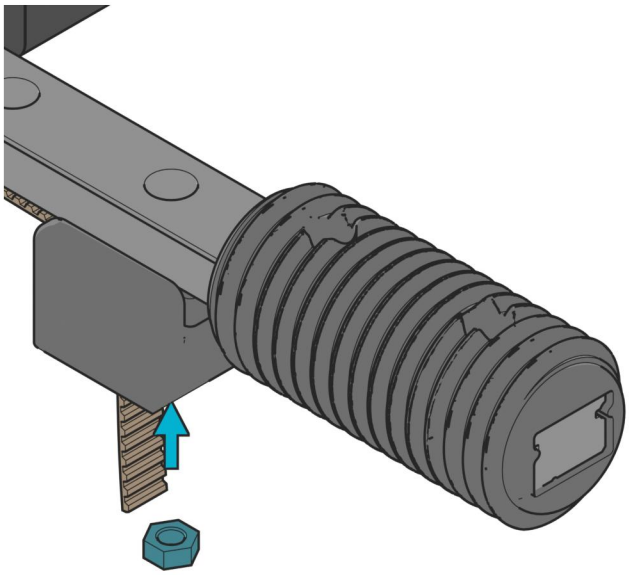


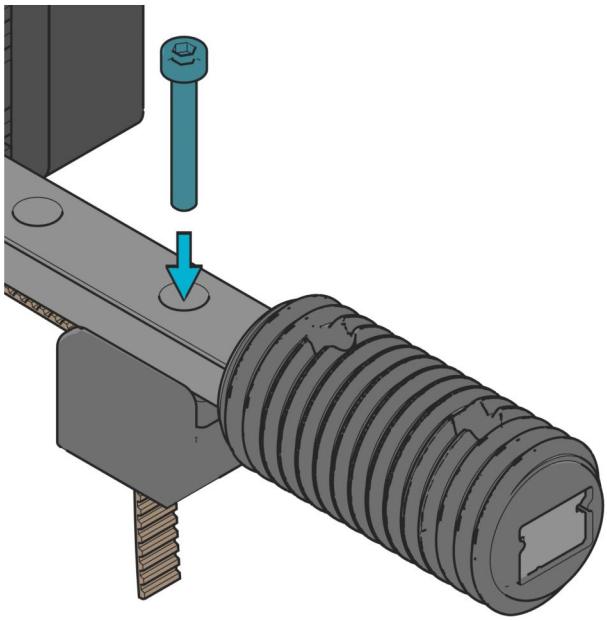
1× M3×20 Screw

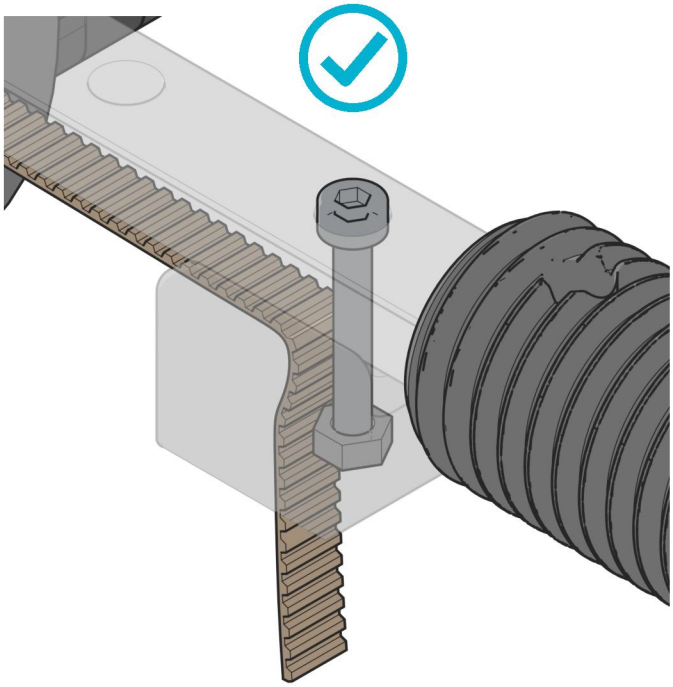
1× M3 Nut

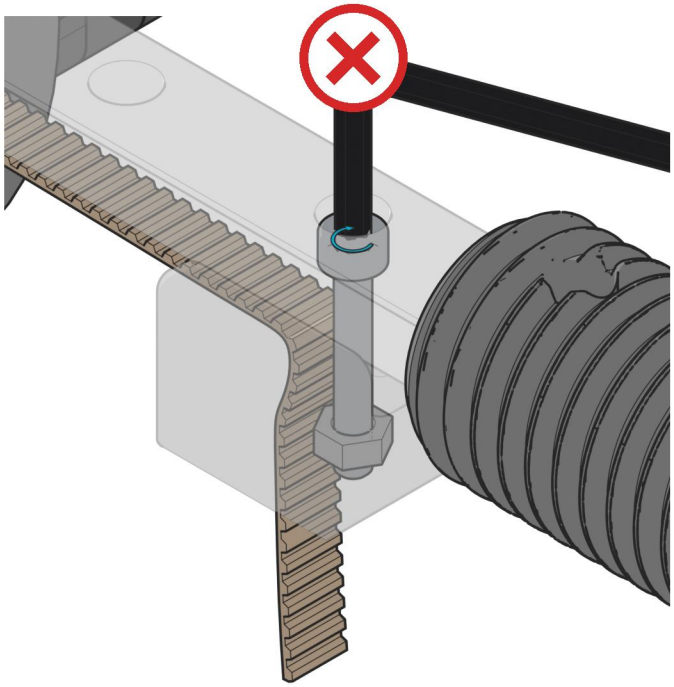


2.5 mm Hex key

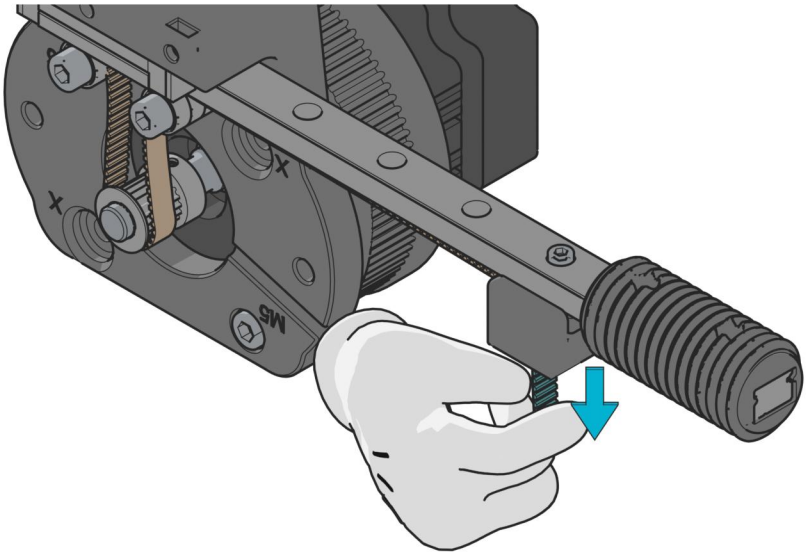


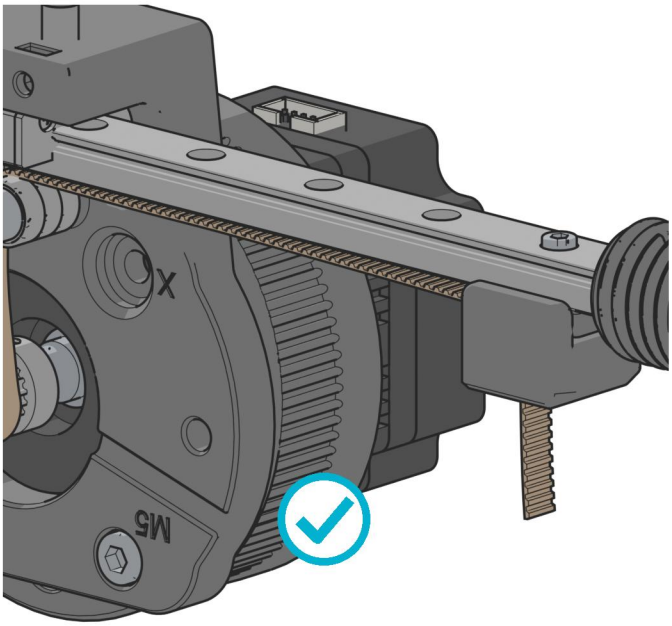


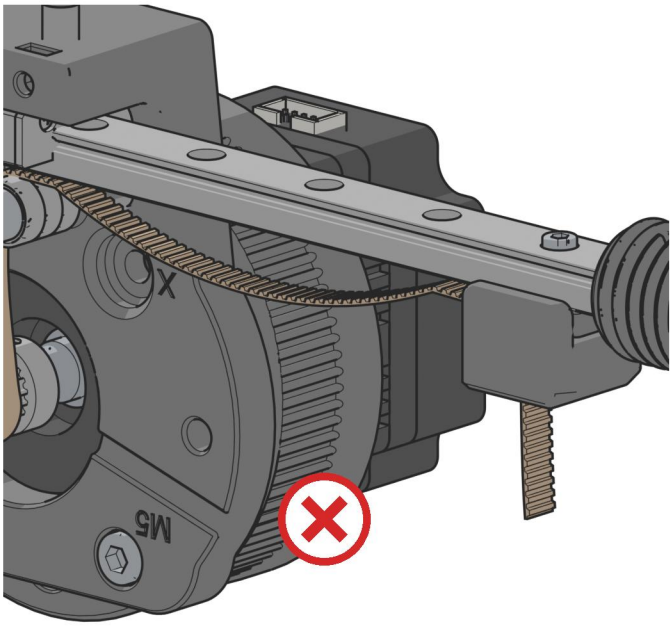




23





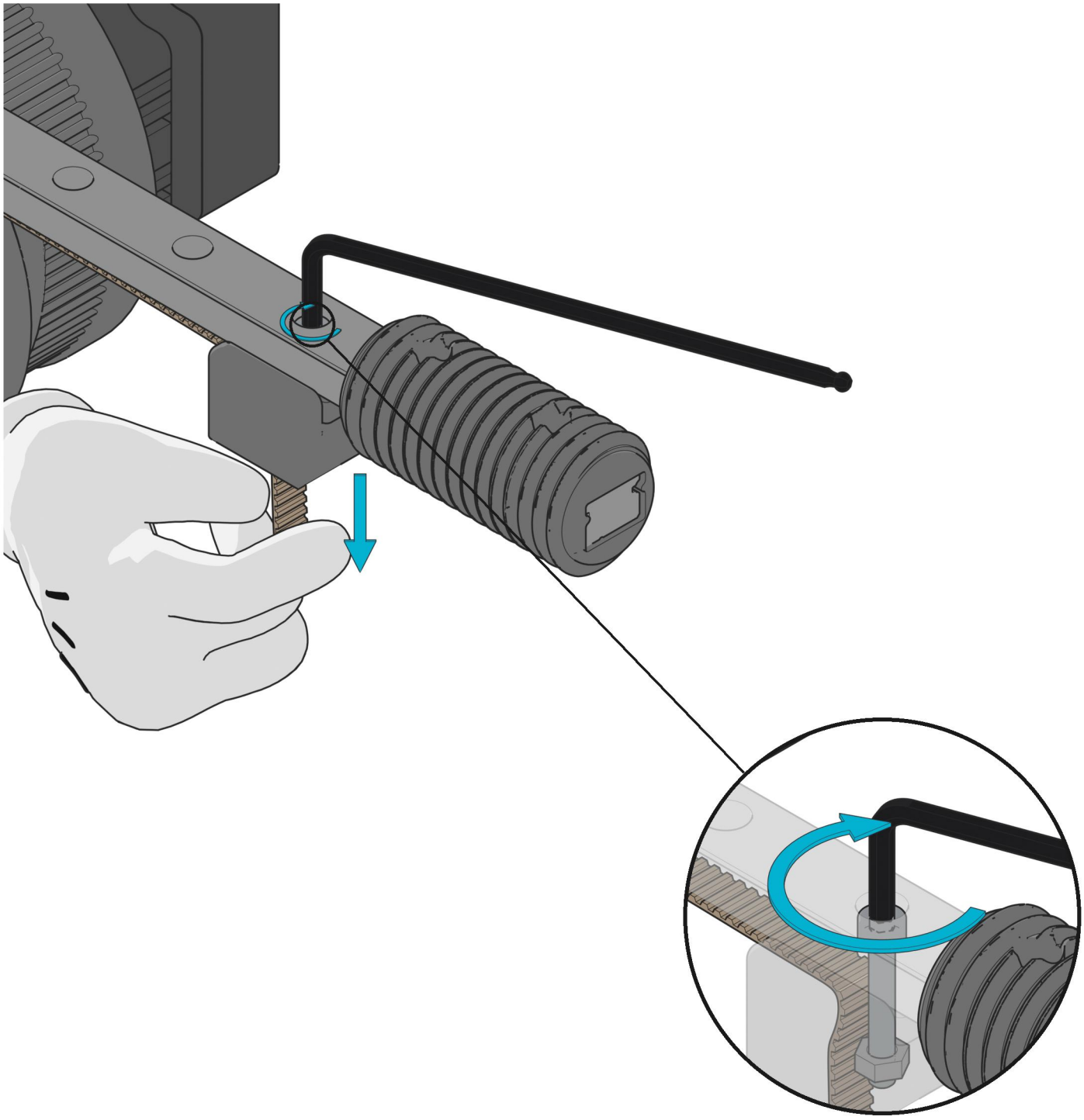


Pull the belt's end down until the belt is taut.

24

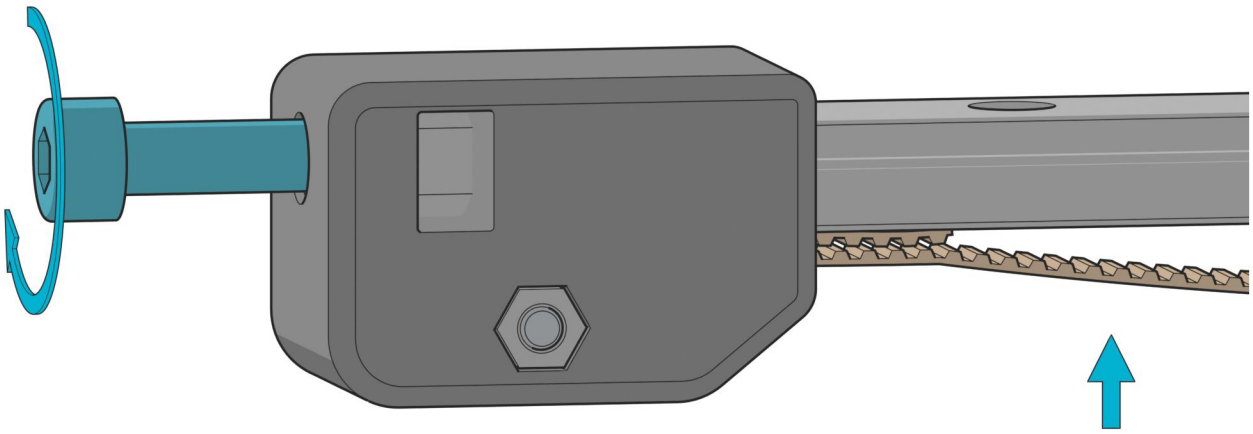


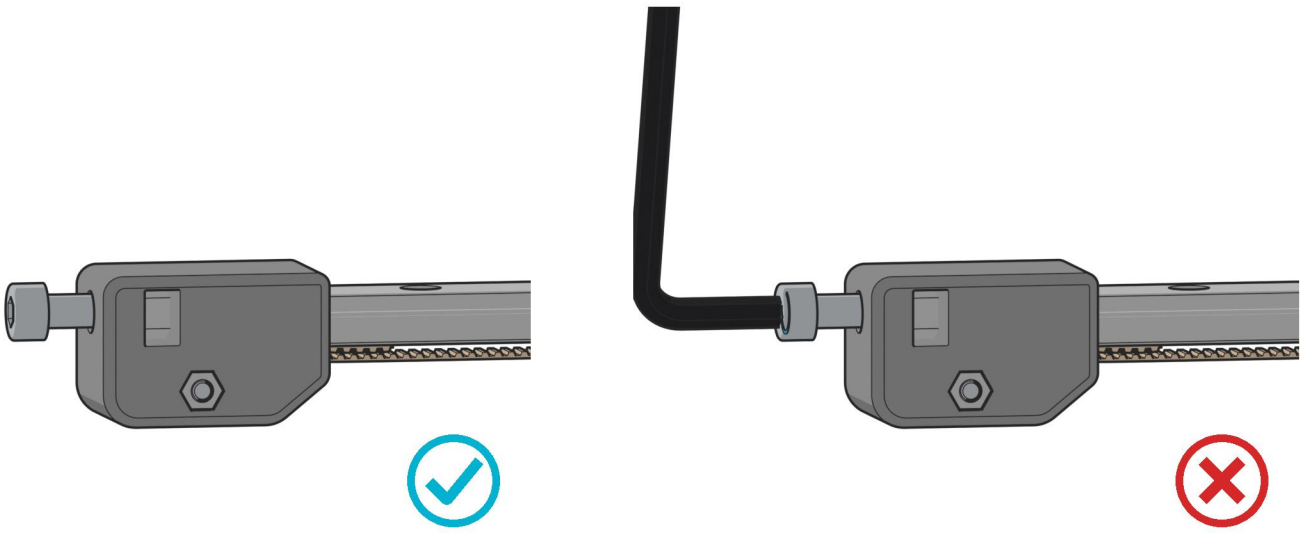
2.5 mm Hex key



Keep pulling the belt taut while you tighten.

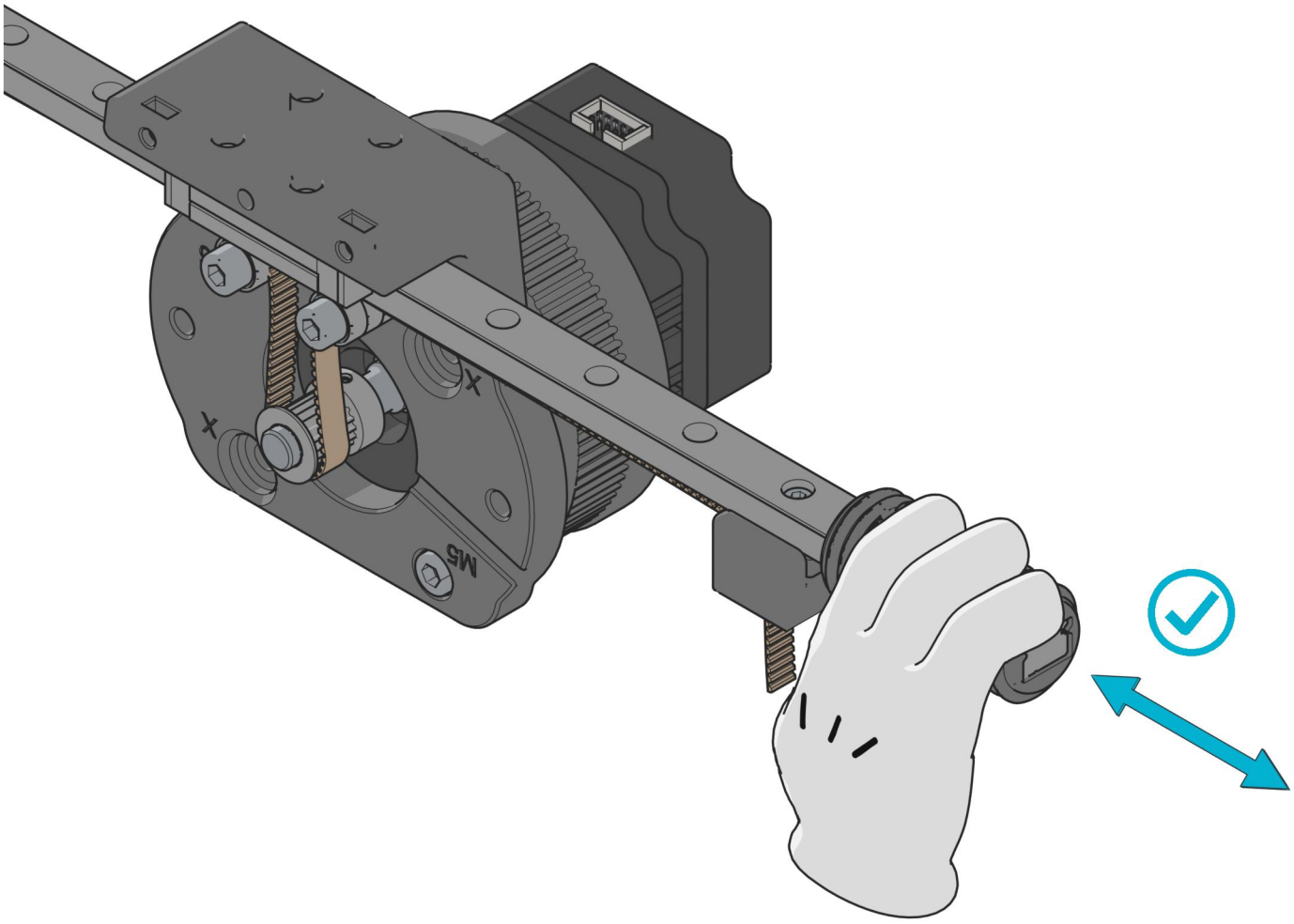
25





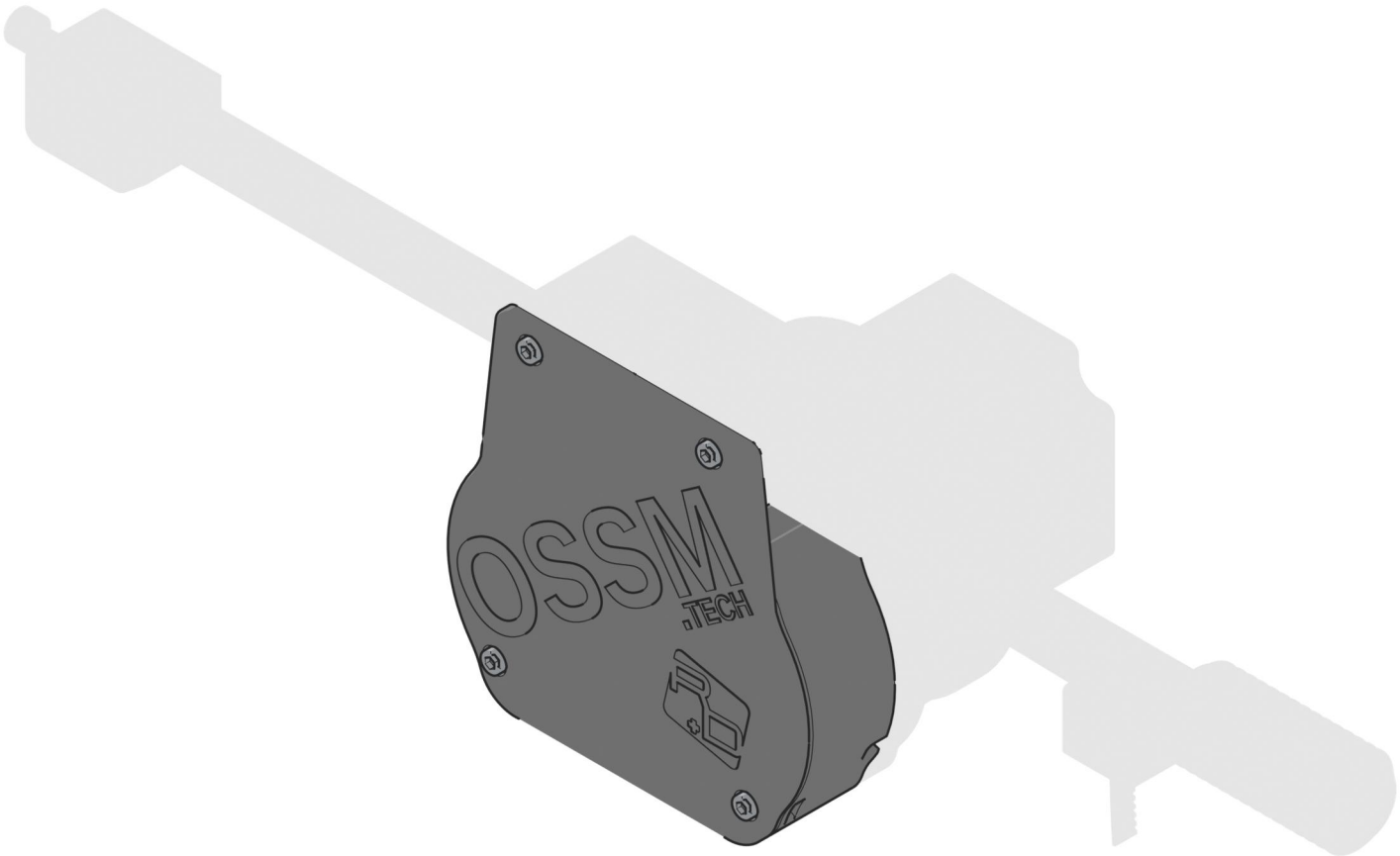
Turn it in by hand: do not over-tighten.

26

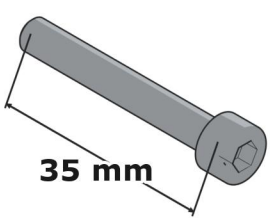


Push and pull the end effector: the rail must slide smoothly through the head.

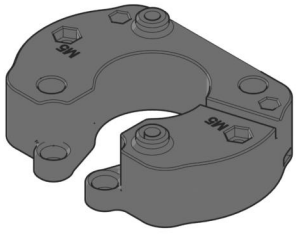
3 Closing the head



27



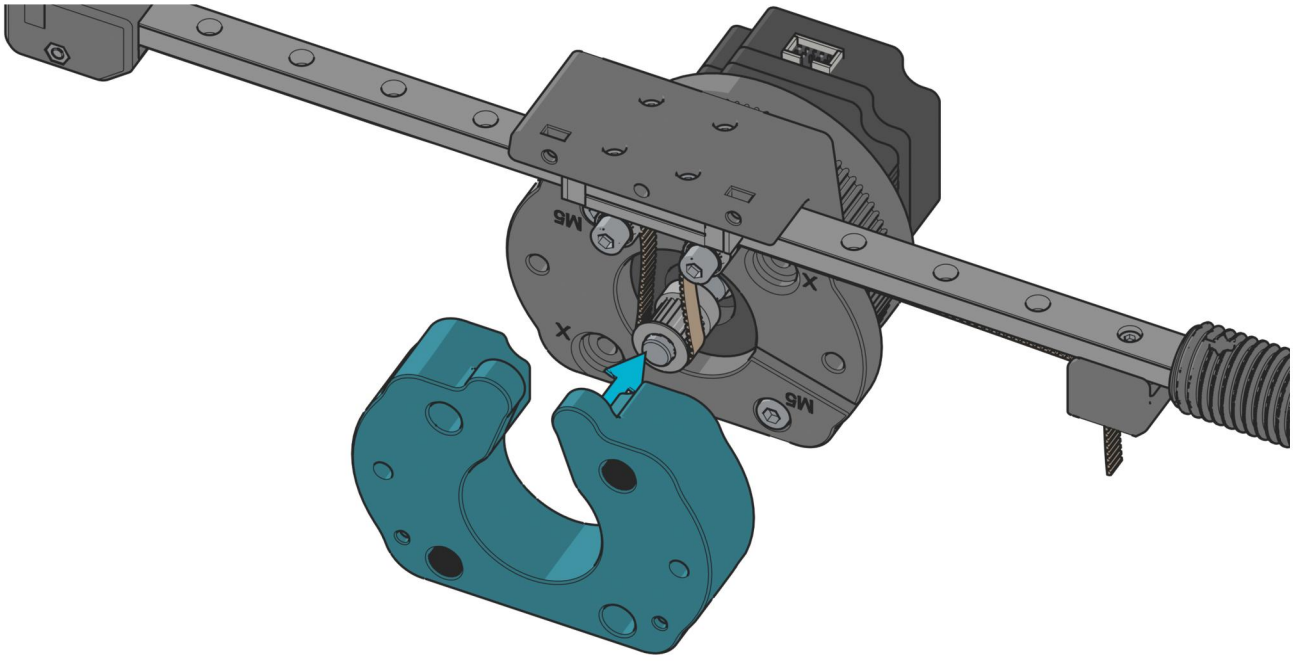
2× M5×35 Screw

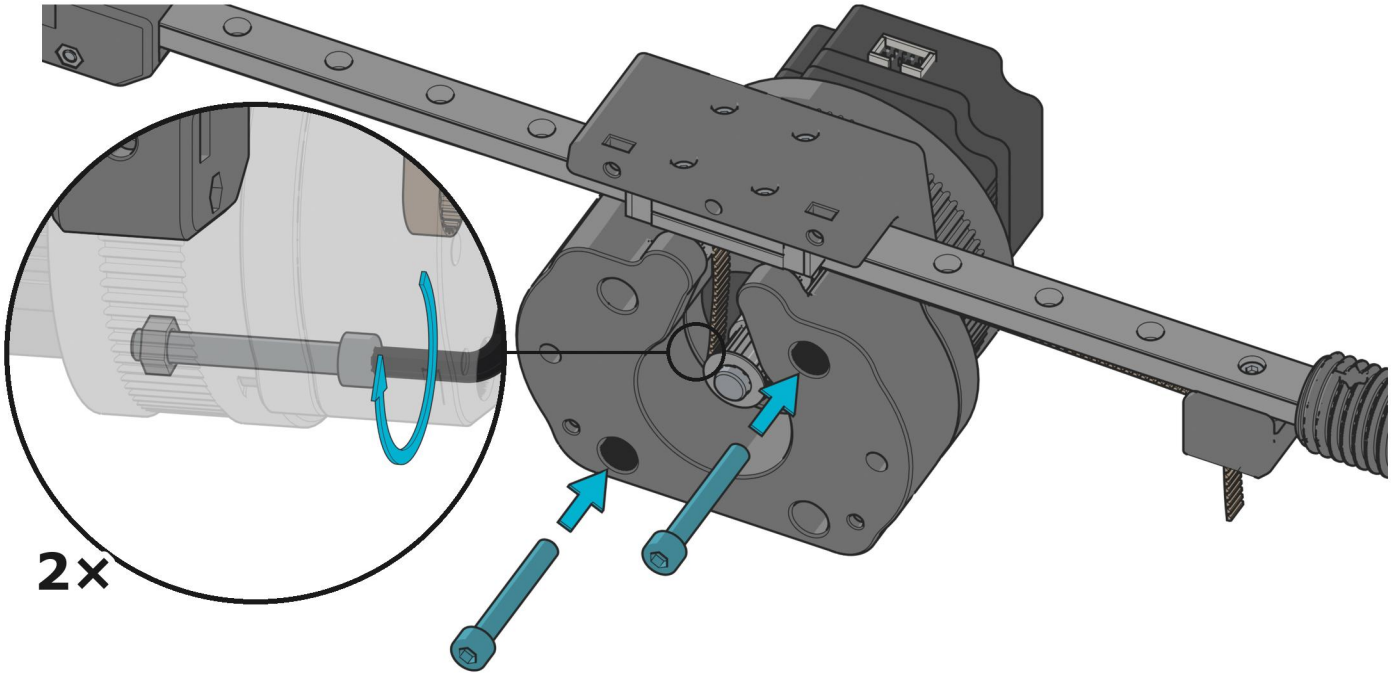


1× Body middle



4 mm Hex key



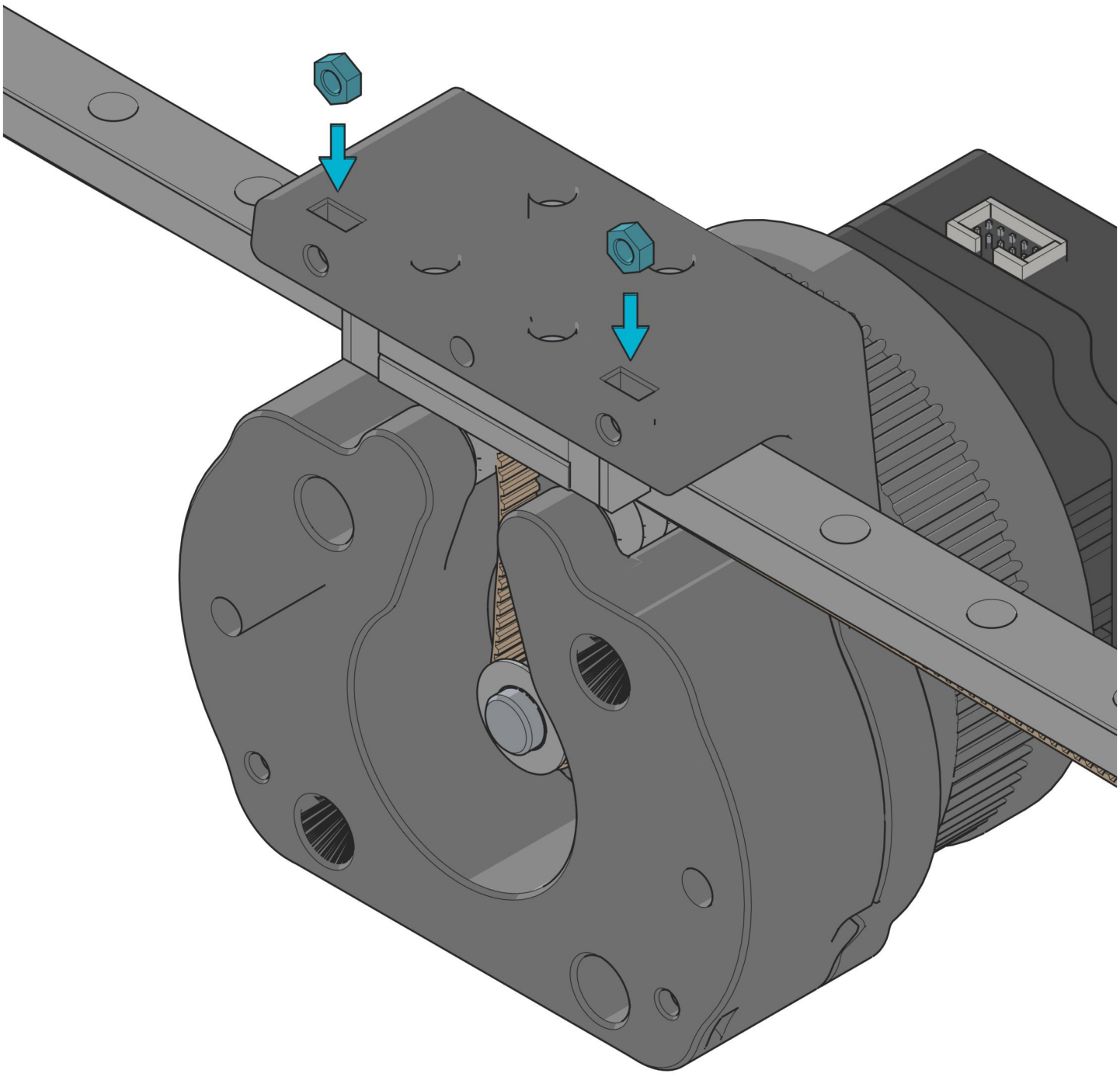


28



2×

M3 Nut

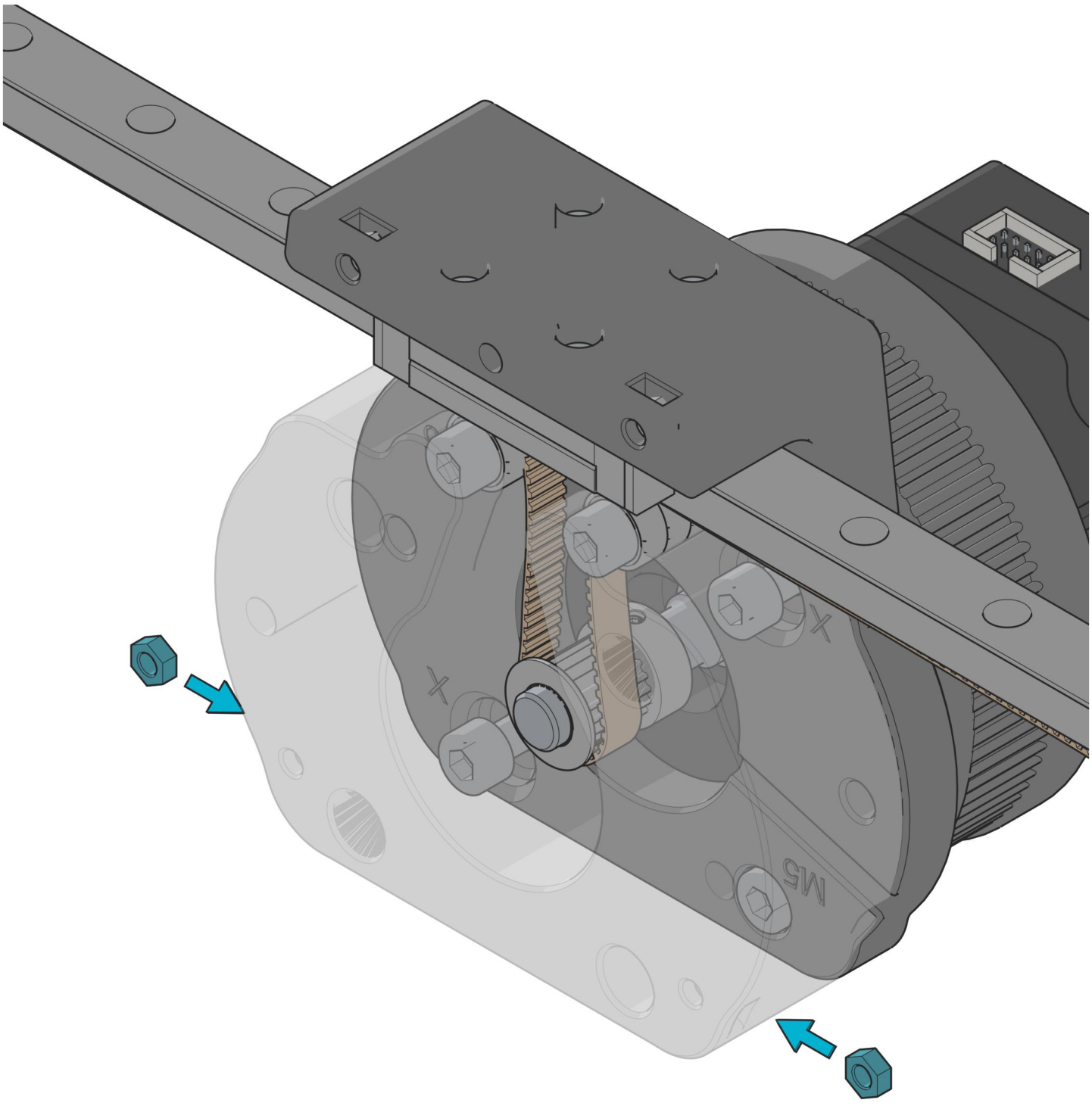


29



2×

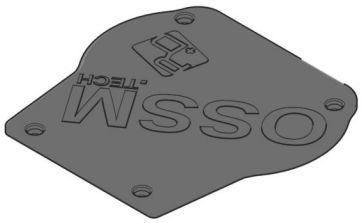
M3 Nut



30



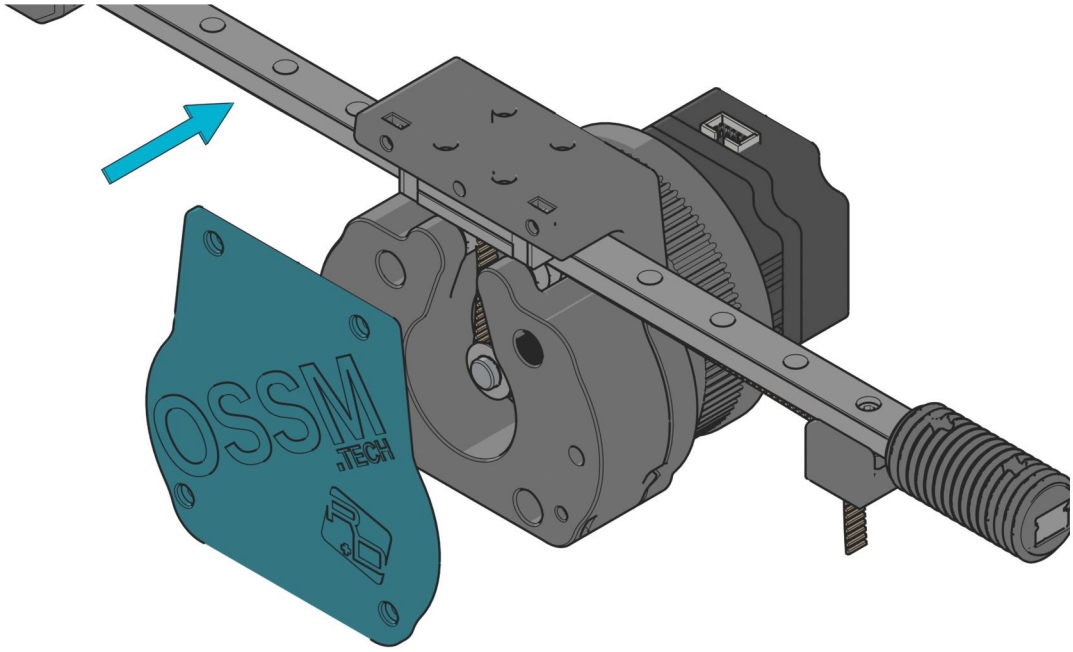
4×
M3×8 Screw

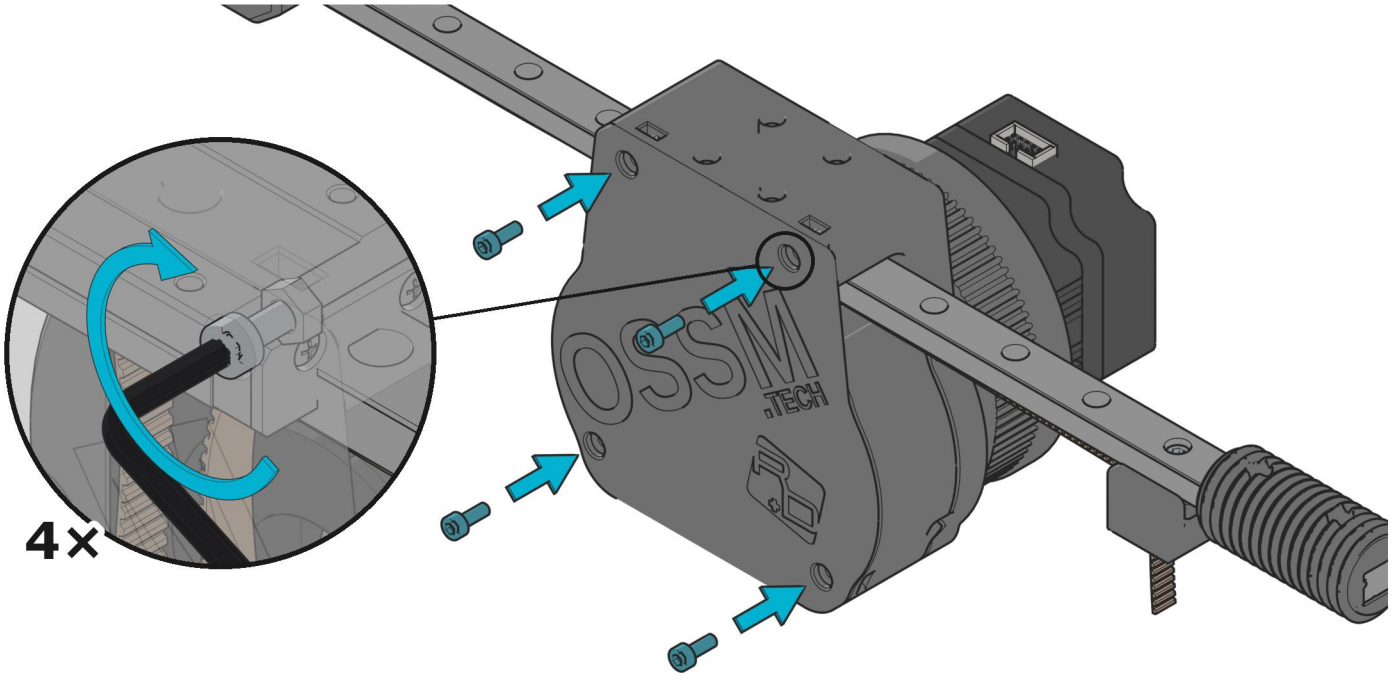


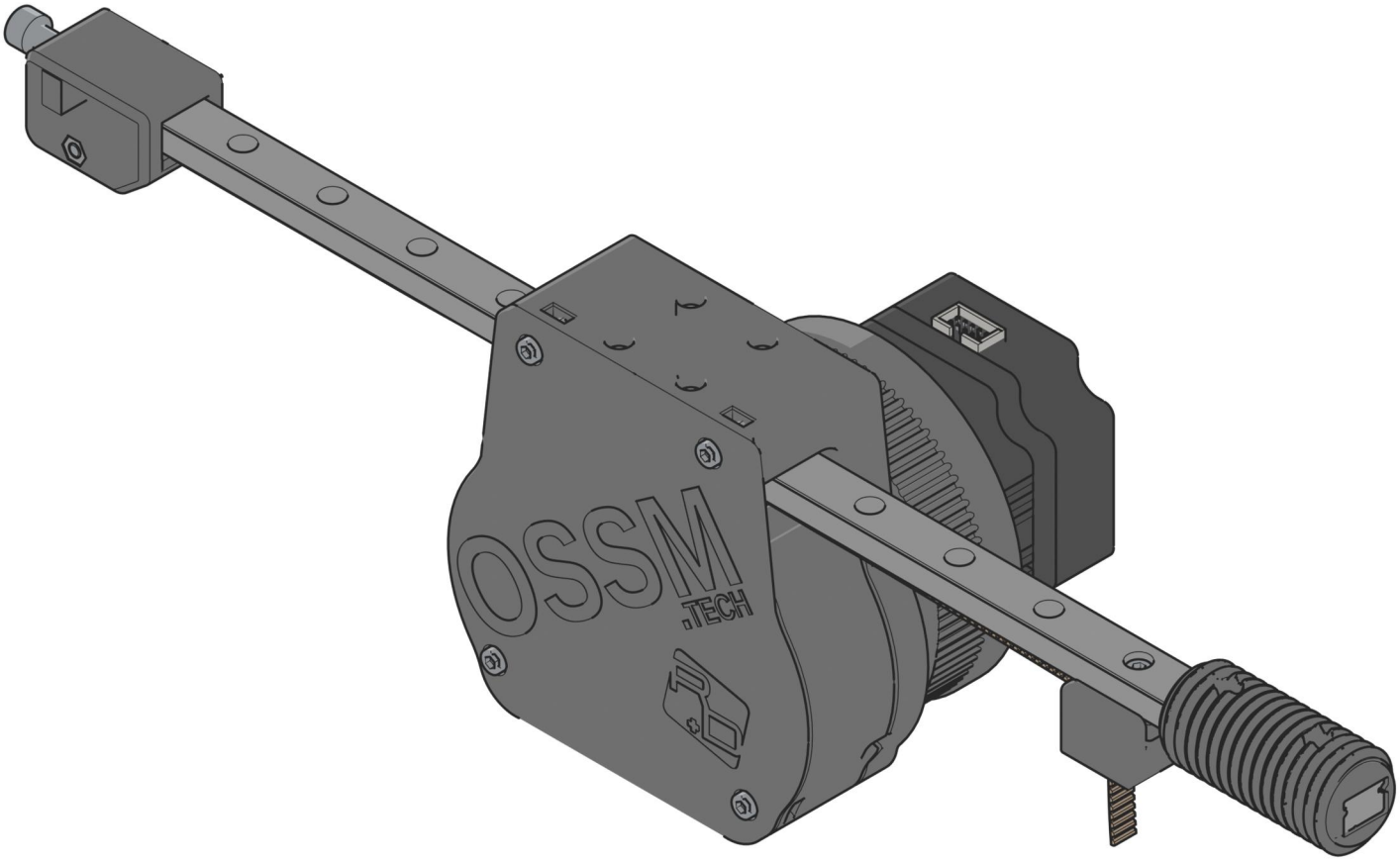
1×
Cover



2.5 mm Hex key







You may now proceed to the electronics and wiring.

RESEARCH + DESIRE

OSSM Actuator Build Guide

Actuator · Version 1.0.17

Support
support@researchanddesire.com
Assembly instructions
ohai.researchanddesire.com/ossm

Gloved hand: "Low Poly Gloved Hand" by CitrusH (@venture.dalton), CC BY 4.0 (creativecommons.org/licenses/by/4.0), posed and restyled: sketchfab.com/3d-models/low-poly-gloved-hand-1f5b60b45c4a43a0984042e23ced7fc6